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How Does Deleveraging Affect Funding Market Liquidity?

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Abstract. How does deleveraging affect the market liquidity of high-embedded-leverage securities issued by financial institutions and the funding constraints of these institutions? We use the forced deleveraging of structured mutual funds during the 2015 Chinese stock market crash to study the effects of deleveraging. Our regression-discontinuity analysis shows that deleveraging significantly reduces the market liquidity of the deleveraging funds' equity units. Moreover, our difference-in-differences analysis shows that deleveraging results in large decreases in subsequent fund flows, stock and cash holdings, and performance, with the impact channeled through the deterioration of the market liquidity of the fund's equity units.

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Keywords: deleveraging • funding market liquidity • funding liquidity crisis

1. Introduction

All over the world, financial institutions are highly leveraged because of the extremely low interest rates in the last decade.¹ Policy makers, academics, and practitioners are concerned about the impact of deleveraging. This paper studies the effect of deleveraging on the market liquidity of high-embedded-leverage securities issued by deleveraging institutions and on the funding liquidity of such institutions in one of the most important capital markets of the world, the Chinese market.

Many investors (e.g., individual investors, mutual funds, pension funds, and even hedge funds) face significant leverage constraints and hence cannot use the desired levels of leverage in their portfolios. Therefore, to achieve high expected returns from using leverage, they have a preference for securities that have high levels of embedded leverage and thus high beta (e.g., leveraged Exchange Traded Funds (ETFs), options, stocks of highly leveraged firms, etc.), and overweigh such securities in their portfolios.² It is known that this investor preference for high-embedded-leverage and high-beta securities increases the prices of such securities, which then leads to low risk-adjusted returns (Baker et al. 2011; Frazzini and Pedersen 2012, 2014).

If, because of operational loss or other funding shocks, issuers of such high-embedded-leverage securities are forced to deleverage to protect debtholder interests, deleveraging can make these originally high-beta

securities much less attractive to leverage-constrained investors. This can result in a substantial reduction in investor demand for these deleveraged securities and thus a significant deterioration of the market liquidity of these securities. The deterioration of the market liquidity of deleveraged securities may in turn amplify the initial funding shocks to the security issuers and further tighten their funding constraints, which can potentially lead to funding liquidity crises. To the best of our knowledge, there is no study in the literature that empirically documents the impact of deleveraging on the market liquidity and funding liquidity of financial institutions.

The lack of empirical evidence on this important issue likely reflects two challenges. First, the leverage of a financial institution is generally unobservable in real time. Second, when financial institutions are forced to deleverage because of an initial trading or operational loss, it is very difficult, if not impossible, to isolate the impact of pure leverage changes from that of the loss. Moreover, other observable or unobservable factors (such as changes in market conditions or implementation of new investment strategies) can lead financial institutions to actively adjust their leverage levels. Because such factors can also affect the market liquidity of securities issued by financial institutions and the funding constraints of these institutions, causality is difficult to establish.

We tackle these two challenges by using the forced deleveraging of some structured mutual funds in the

Chinese market to isolate the effect of deleveraging (i.e., the treatment) on market liquidity, funding constraints, and performance of the treatment funds. Our data set allows us to observe the leverage levels of structured mutual funds at the daily frequency. Because some structured funds crossed the preset leverage limit during our sample period and were subsequently forced to deleverage, these forced deleveraging events provide us an ideal setting to examine the impact of deleveraging. Although fund performance, market conditions, and other factors may influence both fund leverage and equity units' market liquidity, for those fund-day observations with actual leverage near the preset deleveraging threshold, the deleveraging treatment can be viewed as random. We thus use a regression discontinuity design (RDD) to study the impact of deleveraging on the market liquidity of the treatment funds' equity shares.

RDD is a powerful identification strategy. Its premise is that except for the assignment of treatment, which is discontinuous at the deleveraging threshold, the impact of other observable/unobservable factors (e.g., fund trading performance, market conditions, ex ante expected likelihood of forced deleveraging, investor past trading experience and trading performance, etc.) on the market liquidity of a fund's equity shares is similar near the threshold (Lee and Lemieux 2010, Roberts and Whited 2013, Kahraman and Tookes 2017). Hence, by estimating the local average treatment effect around the threshold, we can clearly rule out the influence of other confounding factors and estimate the causal impact of deleveraging on the market liquidity of fund equity shares.

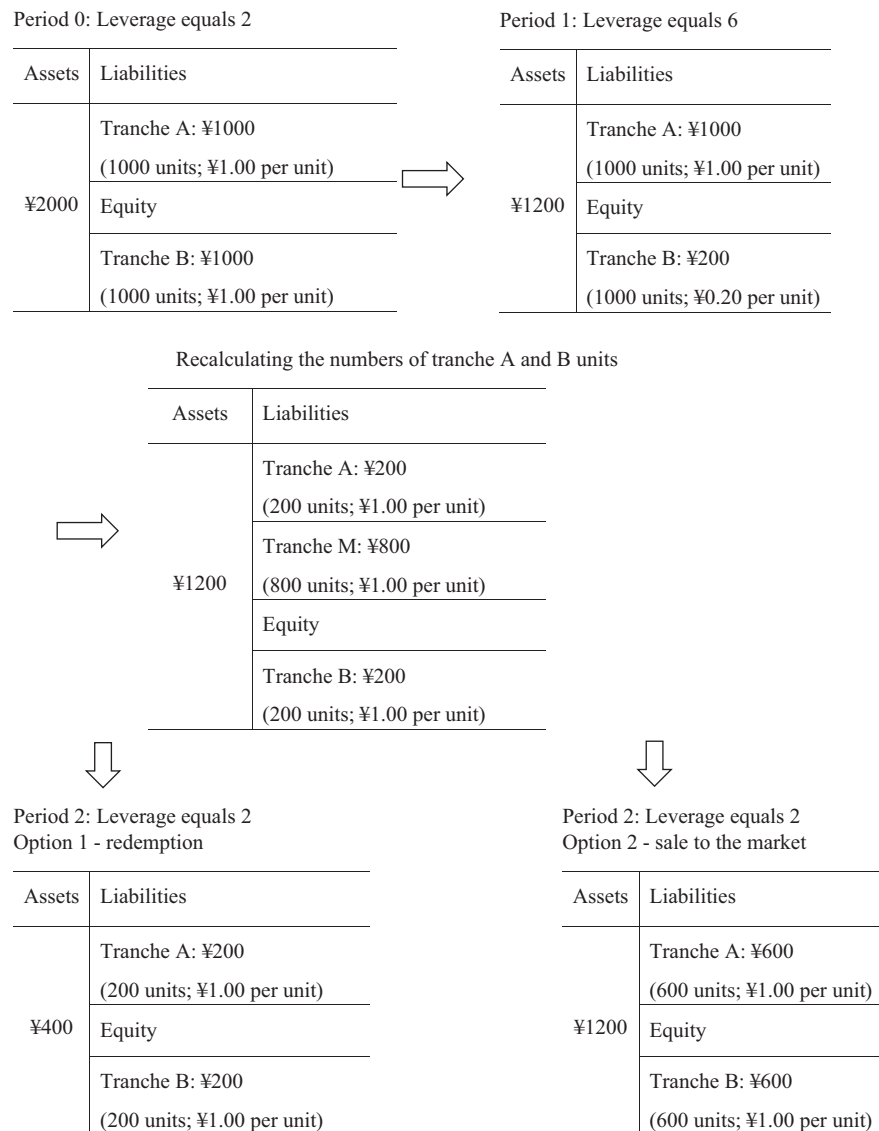
Like other mutual funds in China, structured mutual funds (hereafter, structured funds) raise money from investors by issuing fund units and then using the raised money to invest in stocks and other financial securities such as bonds. Unlike other Chinese mutual funds whose assets-to-equity ratio (i.e., leverage) cannot exceed 1.4, structured funds are allowed to use much higher leverage. What makes structured funds special is that they issue two tranches of fund units. Holders of tranche A debt units receive fixed interest payment, whereas holders of tranche B equity units receive all of the residual income after the fixed interest payment to tranche A debtholders, fund fees, other expenses, and investment losses have been deducted. That is, structured funds effectively borrow from tranche A debtholders to increase the expected returns of their tranche B equity holders.

This high-embedded-leverage feature of tranche B equity units made them very popular among investors who prefer high expected returns. According to the Structured Fund Product Approval Guide issued by the China Securities Regulatory Commission (CSRC) in 2012, when a structured fund is raised, its initial leverage (i.e., the total net asset value of

tranches A and B divided by the net asset value of equity tranche B) should not exceed 2 and subsequent leverage should be maintained below 6 for most funds.³ The leverage limit of structured funds is much higher than the leverage limit of 1.4 for other mutual funds.

When a structured fund is in operation, the value of tranche A debt unit usually changes very little because tranche A debtholders receive fixed interest payment while tranche B equity holders receive the residual income and bear all the investment losses. The fluctuation in actual leverage is mainly driven by changes in the value of tranche B equity unit. Thus, a structured fund typically specifies in its prospectus the threshold net asset value of each tranche B equity unit. Once the fund's actual net asset value per equity unit falls below this deleveraging threshold (e.g., because of trading loss that reduces the net asset value of equity units) and thus the actual leverage level exceeds the preset leverage limit, the fund will be forced to subsequently (i.e., in the next two trading days) bring its leverage down to 2, the initial leverage level. The fund will publicly announce the coming deleveraging.⁴ It will then recalculate/reduce the number of tranche B equity units so that the value of each equity unit is brought back to its initial value. The fund will then correspondingly convert a large portion of tranche A debt units into tranche M parent units to reduce the fund's debt value. These parent units can then be paid back to tranche A debtholders in cash (i.e., redemption) or, if desired by tranche A holders, split equally into tranche A and B units and sold to the market. The purpose of forced deleveraging is to protect tranche A debtholders from investment loss. Because of forced deleveraging protection, there is no instance of tranche A default for structured funds.

Figure 1 illustrates an example of forced deleveraging. There are three periods: 0, 1, and 2. In period 0, a structured fund is set up, and it issues tranche A and B units. Each tranche has 1,000 units, and each unit has an initial value of ¥1.00; thus, the initial leverage is $(1,000 + 1,000)/1,000 = 2$. In period 1, because of investment loss, value of tranche B equity unit falls to ¥0.20 per unit, whereas value of tranche A debt unit remains ¥1.00 (for simplicity). In this case, actual leverage becomes $(1,000 + 200)/200 = 6$, which we assume triggers forced deleveraging. The number of tranche B equity units will then be reduced from 1,000 to 200 so that the value of each B unit is brought back to ¥1.00 (its initial value). The fund also converts 800 tranche A debt units into parent units (tranche M) to reduce the fund's debt value. Thus, the original tranche A debtholders now hold 200 A debt units and 800 tranche M parent units. In period 2, the original tranche A holders can (1) redeem their tranche M parent units and receive cash from the fund or (2) split the 800 tranche M parent units equally into 400

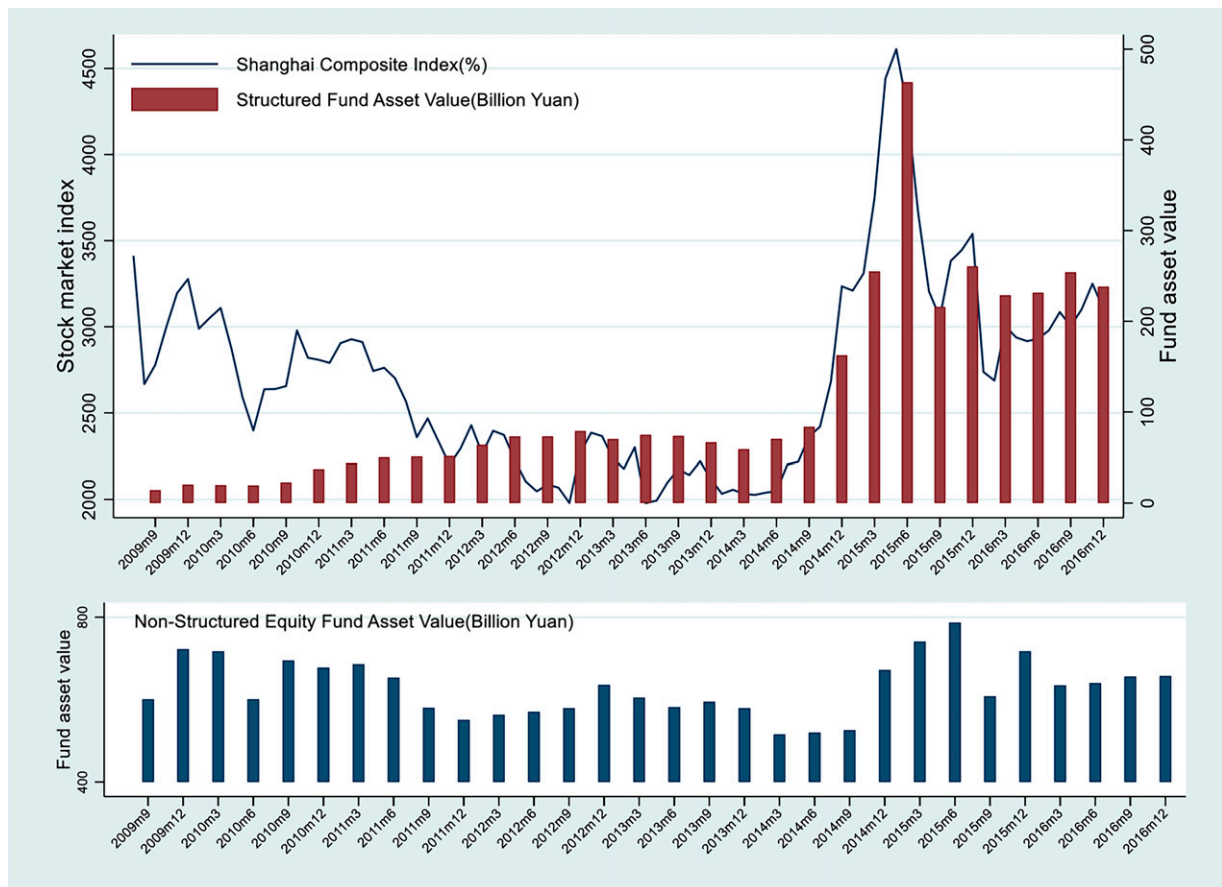
Figure 1. An Illustrative Example of Forced Deleveraging

Notes. This figure gives an illustrative example of forced deleveraging. There are three periods: 0, 1, and 2. In period 0, a structured fund is set up, and it issues tranche A and B units. Each of the A and B tranches has 1,000 units with an initial value of ¥1.00 per unit. Thus, the initial leverage is $(1,000 + 1,000)/1,000 = 2$. In period 1, because of investment loss, value of tranche B equity unit falls to ¥0.20 per unit, while value of tranche A debt unit remains ¥1.00. In this case, actual fund leverage becomes $(1,000 + 200)/200 = 6$, which triggers forced deleveraging. The number of tranche B equity units will then be reduced from 1,000 to 200 so that the value of each B unit is brought back to ¥1.00 (its initial value). The fund also converts 800 tranche A debt units into parent units (tranche M) to reduce the fund's debt value. Thus, the original tranche A holders now hold 200 tranche A debt units and 800 tranche M parent units. In period 2, the original tranche A holders can (1) redeem their tranche M parent units and receive cash from the fund, or (2) split the 800 tranche M parent units equally into 400 tranche A debt units and 400 tranche B equity units and sell the A and B units to the market. Regardless of their choice, actual fund leverage will be brought down to 2. The first option will significantly decrease the AUM of the fund because of investor redemption while the second option will not change the AUM.

tranche A debt units and 400 tranche B equity units and sell the A and B units to the market. Regardless of their choice, actual fund leverage will be brought down to 2. The first option will significantly decrease the assets under management (AUM) of the fund because of investor redemption, whereas the second option will not change AUM. In the online appendix, we describe in detail the actual deleveraging case of the Penghua Gao Tie structured fund.

China's stock market experienced a big crash in mid-2015. The Shanghai Stock Exchange (SSE) composite index steadily increased from 2,038 in the beginning of June 2014 to 5,166 on June 12, 2015. Then, from the next trading day (June 15, 2015), the SSE index started its substantial fall for nearly seven months until January 2016 when the index touched its bottom at 2,655. It then fluctuated at around 3,000 subsequently. Figure 2 shows that during this boom-and-bust

Figure 2. (Color online) Fund Asset Value and Stock Market Index



Notes. The upper figure plots the SSE composite index and the total asset value of structured mutual funds from the third quarter of 2009 through 2016. The height of each bar represents the total asset value of structured mutual funds, shown on the right axis. The line represents the value of the SSE composite index. For comparison, the lower figure plots the total asset value of nonstructured equity mutual funds over the same period.

process, the total AUM of structured funds exponentially increased from ¥70 billion in June 2014 to RMB460 billion in June 2015 (a 657% increase) and then decreased to ¥229 billion in the first quarter of 2016 (a decrease of more than 50%). By contrast, nonstructured equity mutual funds did not experience such dramatic changes in AUM. Structured fund deleveraging did not occur before 2015. However, during the 2015 crash, 78 of the 183 structured funds in our sample crossed the preset leverage limit and were forced to deleverage to the initial leverage level.⁵

We begin our analysis by documenting a positive relation between leverage and the market liquidity of structured funds' equity units before the market crash using ordinary least squares (OLS) panel regressions. Funds with greater leverage have significantly higher order imbalance, narrower effective spread, greater trading volume, and higher turnover for their equity units. This finding is robust to controlling for various fund characteristics, year-day fixed effects, and fund fixed effects, indicating that greater leverage attracts

investors seeking high expected returns. Because the OLS results are prone to potential endogeneity issues such as omitted-variable bias, we next use RDD to better identify the leverage effect on the market liquidity of structured funds' equity units.

We conjecture that when a fund's leverage exceeds the preset limit, the fund's equity units will become much less attractive to these investors (because of the expectation of subsequent deleveraging). The loss of investors' interest and the associated reduction in investor demand will cause a significant deterioration of the equity units' market liquidity. Results from our RDD analysis clearly show a large and statistically significant impact of deleveraging on the market liquidity of a fund's equity units. Compared with the control group (i.e., fund-day observations with just-below-threshold leverage, which have very similar performance and other characteristics as the treatment group), when a treatment fund just crosses the deleveraging threshold (and is thus expected to subsequently deleverage to its initial leverage in the next two trading days), its order

imbalance decreases on average 2.1 times, effective spread increases 5.8 times, and trading volume decreases by 43%, relative to their respective sample standard deviations. These results lend strong support to our conjecture. Additional empirical evidence further rules out several alternative explanations and suggests that deleveraging negatively impacts the market liquidity of treatment funds' equity units due to investors' preferences for high-embedded-leverage securities.

Deleveraging results in a large portion of tranche A debt units being converted into parent units (tranche M units). The latter can then be paid back to tranche A debtholders in cash (i.e., redemption) or, if desired by tranche A holders, split equally into tranche A and B units and sold to the market.⁶ Because of the significant reduction in the market liquidity of its tranche B equity units (e.g., high effective spread and low trading volume) after a fund crosses the deleveraging threshold, most, if not all, tranche A investors will go for redeeming their investment and demanding cash from the treatment fund (rather than splitting their M units into A and B units and sell them to the market at a discount due to the low B unit market liquidity). Moreover, because of the low embedded leverage of the B units after deleveraging, many investors of B units no longer want to hold these securities and, because of the deteriorated B unit market liquidity, may also choose to exit through redemption. Significant redemption pressures can lead to funding liquidity crises of these treatment funds. These redemption pressures can cause the treatment funds to have a fire sale of their stock holdings and produce low relative fund performance.

We use data on quarterly fund flows and portfolio holdings around the deleveraging events to examine the impact of deleveraging on the funding constraints of the treatment funds. We use the event-window cohort approach in a difference-in-differences (DID) regression framework, as suggested by Gormley and Matsa (2011).⁷ A potential concern for the DID analysis is that the treatment (i.e., deleveraging) may be driven by recent fund performance and/or correlate with other fund characteristics and hence may not be random. To alleviate this concern, we use propensity score matching (PSM) to select the control group (i.e., fund-quarter observations of structured funds that did not deleverage during the sample period) that has no observable difference in ex ante fund characteristics compared with the treatment group. We then construct event-window cohorts from the PSM sample for our DID analysis.⁸ The results clearly show that, compared with structured funds that did not deleverage, the treatment funds saw large redemption outflows and large reductions in stock holdings, cash holdings, and fund returns in subsequent quarters. These results indicate that these funds experienced severe funding liquidity crises.

We further examine whether the large reduction in market liquidity of their equity units after crossing the deleveraging thresholds causes the treatment funds' funding liquidity crises. Because of this significant reduction in market liquidity, investors choose to redeem their investment rather than to sell their fund units at a discount to the market. Our results show that the larger the reduction in the market liquidity of the equity units around deleveraging is, the greater the impact of deleveraging on subsequent outflows, stock holdings, cash holdings, and fund returns. Thus, our empirical evidence clearly suggests that deleveraging results in a loss of leverage-constrained investors' interests and demand of the treatment funds' equity units and hence significantly decreased the market liquidity of the equity units, which in turn leads investors to redeem their investment from these funds, thus causing their funding liquidity crises.

The rest of the paper proceeds as follows. Section 2 discusses the related literature and develops the hypotheses. Section 3 describes the data and presents sample descriptive statistics. Section 4 documents the relation between structured fund leverage and the market liquidity of equity units before the market crash. Section 5 provides the empirical results from the RDD analysis on the impact of deleveraging on the market liquidity of the treatment funds' equity units. Section 6 presents the empirical results from the DID analysis on the impact of deleveraging on the treatment funds' fund flows, stock and cash holdings, and fund performance. Section 7 shows that the impact of deleveraging on funding liquidity is channeled through the deterioration of equity units' market liquidity. Section 8 concludes. The Appendix provides the definitions of the variables used in the study (Table A.1) and additional empirical evidence. The online appendix describes the deleveraging case of the Penghua Gao Tie structured fund.

2. Literature Review and Hypothesis Development

2.1. Literature Review

Our study is related to the literature on investors' leverage constraints and preference for high-beta securities (Barber and Odean 2000; Baker et al. 2011; Frazzini and Pedersen, 2012, 2014; Christoffersen and Simutin 2017). For example, Barber and Odean (2000) show individual investors prefer high beta stocks in their common stock investment. Baker et al. (2011) point out that high-beta and high-volatility stocks have long underperformed compared with low-beta and low-volatility stocks and attribute this underperformance to the mandate of institutional investors to beat investment benchmarks. Frazzini and Pedersen (2014) develop a dynamic model of leverage constraints and show that leverage-constrained investors prefer stocks with high

embedded leverage and bid up the prices of high-beta stocks, which lead to low risk-adjusted returns for such stocks.⁹ Christoffersen and Simutin (2017) show evidence that mutual funds with large defined-contribution pension assets tilt their portfolios toward high-beta stocks to beat benchmarks. We contribute to this literature by documenting novel evidence of a (potentially) causal and negative impact of deleveraging on the market liquidity of high-embedded-leverage securities issued by financial institutions and showing how this market liquidity deterioration can translate into funding liquidity crises for the deleveraging institutions.

Our study is also related to the literature on investor coordination failures and funding liquidity crises (Diamond and Dybvig 1983, Chen et al. 2010, He and Xiong 2012, Brunnermeier and Oehmke 2013, Kacperczyk and Schnabl 2013). This literature suggests that when financial institutions suffer operational losses, investor runs and funding liquidity crises can occur as a result of investors trying to achieve first-mover advantage in redeeming their assets. We contribute to this literature by showing that deleveraging and its negative impact on the market liquidity of high-embedded-leverage securities can trigger the funding liquidity crises of deleveraging financial institutions.

Finally, our study is related to the literature on the interaction between *traders'* funding liquidity and asset market liquidity. In their seminal work, Brunnermeier and Pedersen (2009) predict that initial funding-liquidity shocks can lead traders to deleverage by fire selling assets that in turn depresses asset prices and decreases asset market liquidity. Deteriorated asset market liquidity further tightens traders' funding constraints and forces them to sell even more assets, leading to a downward liquidity spiral and cross-trader contagion.¹⁰ Empirical evidence from the U.S. stock market (Hameed et al. 2010), hedge funds (Teo 2011), repo market (Gorton and Metrick 2012), and Indian stock market (Kahraman and Tookes 2017) support these theoretical predictions. Moreover, consistent with the theoretical predictions, Bian et al. (2018) show that during the 2015 Chinese stock market crash, selling pressures because of margin traders' deleveraging further depress stock prices, whereas Hu et al. (2019) show that margin traders' deleveraging results in spillover of stock market illiquidity within the same industry and across different industries. Bian et al. (2019) further identify portfolios of margin traders as a source of market contagion.

Different from these studies that focus on the interaction between traders' funding liquidity and asset market liquidity, we document a novel interaction between financial institutions' funding liquidity and the market liquidity of high-embedded-leverage securities: Deleveraging makes the originally high-embedded-leverage securities issued by financial institutions much less attractive to investors seeking high expected returns and causes a significant deterioration of market liquidity of

such securities; this market liquidity deterioration in turn further tightens the funding constraints of the deleveraging institutions, leading to funding liquidity crises.

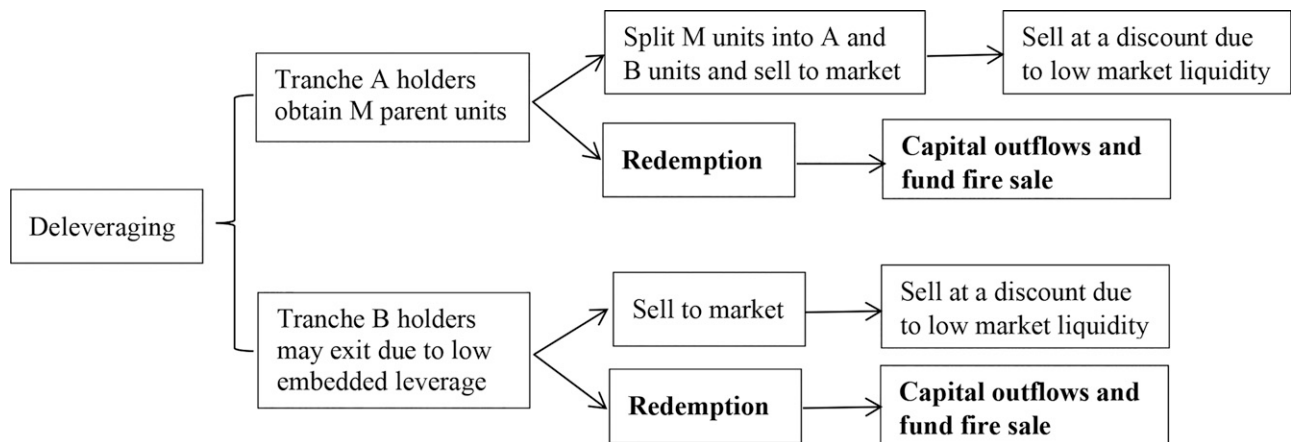
2.2. Hypothesis Development

As mentioned earlier, many investors face significant leverage constraints. To achieve high expected returns, they thus prefer financial securities with high embedded leverage (Frazzini and Pedersen 2014). Equity units of structured funds are particularly popular to such leverage-constrained investors because of their high embedded leverage.¹¹ Tranche B equity holders effectively borrow from tranche A debtholders for investment purposes and receive the benefits of high expected returns because of high leverage. If a structured fund crosses the preset deleveraging threshold because of the fund's trading losses, its tranche B equity units will become much less appealing to leverage-constrained investors because investors expect the fund to deleverage subsequently. The loss of investors' interest can result in a substantial reduction in the demand for the fund's equity units, which will in turn lead to significant deterioration of the equity units' market liquidity. By contrast, even if the fund's actual leverage approaches the limit for deleveraging, demand for the equity units may still be relatively high because (1) higher leverage can provide greater benefits to leverage-constrained investors who seek high expected returns, and (2) the price of tranche B equity units may be relatively low when the fund approaches the deleveraging threshold (i.e., because of trading losses). We thus have the following hypothesis.

Hypothesis 1. *Deleveraging causes a significant deterioration in the market liquidity of the (deleveraging) treatment fund's equity units.*

As demonstrated in Figure 3, there are two reasons why the significant deterioration of the equity units' market liquidity will increase the subsequent redemption pressures on the treatment funds.

First, because of the significantly deteriorated market liquidity of the equity units, most of the original tranche A debtholders will likely choose to redeem their parent units rather than split them equally into debt and equity units and then sell them at discounted prices. Second, the significantly decreased fund leverage makes the tranche B equity units much less appealing to tranche B investors. Some of these investors no longer want to hold the equity units and may also want to exit through redeeming their investment from the treatment fund (rather than selling their B units to the market at discount prices due to the low market liquidity).¹² Consequently, the fund may experience large capital outflows and fall into a funding liquidity crisis subsequently. The treatment fund may need to draw down its cash holdings to meet the significantly

Figure 3. Deterioration of the Equity Units' Market Liquidity and Redemption Pressures

Notes. This figure illustrates why the significant deterioration of the equity units' market liquidity will increase the redemption pressures on the treatment fund. First, because of the significantly deteriorated market liquidity of the B equity units, most of the original tranche A debtholders will likely choose to redeem their M parent units (obtained through deleveraging) rather than split them equally into A and B units and then sell them at discounted prices. Second, the significantly decreased fund leverage and deteriorated market liquidity make the tranche B equity units much less appealing to tranche B investors. Some of these investors no longer want to hold the equity units and may also exit through redeeming their investment from the treatment fund (rather than selling their B units to the market at a discount due to the low market liquidity). Consequently, the fund may experience large capital outflows. The treatment fund may need to draw down its cash holdings to meet the significantly increased redemption demand. It may even need to do a fire sale of its stock holdings to meet the redemption demand, which can lead to significantly decreased fund performance.

increased redemption demand. It may even need to have a fire sale of its stock holdings to meet the redemption demand, which can lead to significantly decreased fund performance.

Hypothesis 2. *Compared with structured funds that do not deleverage, treatment funds subsequently experience large fund outflows and large reductions in stock and cash holdings and fund performance.*

We conjecture that the funding liquidity crises of the treatment funds are caused by the large reduction in the market liquidity of their tranche B equity units after crossing the deleveraging thresholds. The larger the reduction in the market liquidity, the greater the incentive for investors to redeem their investment rather than to sell their fund units to the market at discounted prices, thus resulting in larger fund outflows and greater reduction in fund stock holdings, cash holdings, and fund performance.

Hypothesis 3. *The impact of deleveraging on fund flows, stock holdings, cash holdings, and fund returns is channeled through the deterioration of the market liquidity of the equity units.*

3. Data and Sample Descriptive Statistics

We obtain the data of the Chinese structured fund from the China Stock Market & Accounting Research (CSMAR) database, a leading Chinese stock market and financial data vendor. In total, there are 183 structured funds in our sample. The sample period is from June 1,

2014 to December 31, 2016, which covers the recent boom–bust cycle of the Chinese stock market. The CSMAR structured fund database provides daily data on fund AUM, fund-unit net asset values, unit prices, and unit returns. The CSMAR mutual fund database provides quarterly data on fund flows, portfolio holdings, cash holdings, and fund institutional ownership of all mutual funds including structured funds. To construct the measures of market liquidity, we obtain the high-frequency trades and quotes data for structured fund units from Great Fortune Data Center (<http://www.licai668.cn/>), a professional Chinese high-frequency data vendor.

Structured funds typically specify in their respective prospectus a threshold (minimum) net asset value of tranche B equity unit that triggers deleveraging. We are able to identify the threshold net asset value of tranche B equity unit for 142 of the 183 structured funds from fund prospectus and CSMAR. We identify the threshold net asset value of tranche B equity unit for the rest of the structured funds according to the Structured Fund Product Approval Guide issued by the CSRC. Table A.2 in the Appendix reports the leverage threshold distribution of the structured funds according to the threshold net asset value of tranche B equity unit that triggers deleveraging. The vast majority (more than 70%) of structured funds specify the threshold net asset value of tranche B equity units such that the preset leverage limit is five, whereas other structured funds allow the leverage limit to be higher or lower than five. In subsequent analysis, we show that excluding these latter funds from the sample does not change our main findings qualitatively.

As mentioned previously, a forced deleveraging event is triggered when the net asset value of a fund's tranche B equity unit falls below the specified threshold value (and thus the actual fund leverage exceeds the preset limit). When this happens, the fund must recalculate/reduce its number of outstanding tranche A and B units in the next two trading days to bring its leverage down to the initial leverage level. We manually identify all such deleveraging events from the public announcements of the structured funds over the sample period.¹³ In total, 78 structured funds deleveraged during our sample period.¹⁴ All of these deleveraging events occurred after the debut of the stock market crash on June 12, 2015. For each deleveraging event, we manually check the fund leverage level before and after the event. The leverage level crossed the preset limit on the event day and remained high on the first trading day after the event day for all treatment funds. On the second trading day after the event day, the numbers of outstanding tranche A and B units were recalculated/reduced: the leverage level significantly decreased to about two for all treatment funds. That is, the leverage levels of all the treatment funds indeed significantly decreased to the initial leverage within two trading days after crossing the preset deleveraging thresholds.

We are interested in studying the impact of deleveraging on the market liquidity of the fund's equity units. We use four daily measures of market liquidity. The first measure, order imbalance, *OI*, is calculated as follows:

$$OI_{it} = \frac{Buy\ orders_{it} - Sell\ orders_{it}}{0.5 \times (Buy\ orders_{it} + Sell\ orders_{it})}. \quad (1)$$

We use the tick-by-tick trading data of the structured fund's tranche B equity units to calculate the sum of all *buy* orders and the sum of all *sell* orders for each fund *i* in each trading day *t*. *OI* is then calculated for each fund-day using Equation (1).

The second measure, effective spread, *ES*, is calculated as follows:

$$ES_{it} = \frac{100 \times |Transaction\ Price_{it} - 0.5 \times (Bid_{it} + Ask_{it})| \times 2}{0.5 \times (Bid_{it} + Ask_{it})}. \quad (2)$$

Compared with quoted spread, effective spread better reflects the actual trading costs in the market (Bessembinder and Venkataraman 2010, Kahraman and Tookes 2017). To calculate *ES* for a fund's equity units in a trading day, we first calculate the effective spread of each trade during the day using Equation (2) and then average all the effective spreads across the day for that fund.

The third measure, *Ln(Vol)*, is natural logarithm of the fund's tranche B daily trading volume. The fourth measure, *Turnover*, is daily turnover calculated as daily trading volume divided by the total number of

tranche B equity units that are outstanding for the fund. The market liquidity of a fund's tranche B reflects investors' ability to quickly sell the fund's tranche B equity units without having to significantly reduce the selling price. Thus, the greater the *OI* and the smaller the *ES*, the greater is the market liquidity of the fund's equity units. *Ln(Vol)* and *Turnover* capture investors' activeness in trading a fund's tranche B equity units, which is likely driven by their heterogeneous beliefs on future returns (Xiong and Yu 2011, Hong and Sraer 2013). The higher *Ln(Vol)* and *Turnover* are, the greater is the market liquidity of the fund's equity units.

Table 1 shows the sample descriptive statistics. Panel A provides the means and medians (in parentheses) of the time-varying variables for the deleveraging funds and the nondeleveraging funds respectively, both before and after June 12, 2015. We also provide the differences in means and medians between the two groups of funds. Panel B provides the means and medians (in parentheses) of the time-invariant variables for the deleveraging funds and the nondeleveraging funds, respectively, as well as the differences in means and medians between the two groups of funds. Table A.1 in the Appendix provides the detailed variable definitions.

The funds' average daily leverage is 2.51 (1.94) for the nondeleveraging funds (the deleveraging funds) in the subperiod from June 1, 2014 to June 12, 2015; it is 2.57 (2.26) for the nondeleveraging funds (the deleveraging funds) in the subperiod after June 12, 2015. The average distance of the fund's actual leverage to its deleveraging threshold in the 90 trading days immediately before June 12, 2015, *LD_Before_MktCrash*, is 3.57 (3.04) for the nondeleveraging funds (the deleveraging funds). The average daily order imbalance, *OI*, is -0.12 and -0.10 (-0.12 and -0.07) for the tranche B equity units of the nondeleveraging funds and the deleveraging funds, respectively, before (after) June 12, 2015. The average daily effective spread, *ES*, is 3.38 and 4.31 (2.99 and 5.08) for the equity units of the nondeleveraging funds and the deleveraging funds, respectively, before (after) June 12, 2015. The average daily log trading volume, *Ln(Vol)*, is 13.73 and 15.81 (14.15 and 15.28) for the equity units of the nondeleveraging funds and the deleveraging funds, respectively, before (after) June 12, 2015. The average daily *Turnover* is 0.10 and 0.20 (0.07 and 0.13) for the equity units of the nondeleveraging funds and the deleveraging funds, respectively, before (after) June 12, 2015.

In the subperiod from June 1, 2014 to June 12, 2015 (the subperiod before the crash), compared with the nondeleveraging funds, the deleveraging funds on average have lower leverage (*Leverage*), higher order imbalance (*OI*), higher effective spread (*ES*), higher daily

Table 1. Descriptive Statistics

Panel A							
Variables	Number of observations	Before June 12, 2015			After June 12, 2015		
		Nondeleveraging	Deleveraging	Difference	Nondeleveraging	Deleveraging	Difference
Daily frequency							
Leverage	75,145	2.512 (2.238)	1.936 (1.819)	0.576*** (0.419***)	2.565 (2.459)	2.258 (2.031)	0.307*** (0.428***)
OI	71,995	−0.116 (−0.107)	−0.098 (−0.072)	−0.018*** (−0.035***)	−0.12 (−0.086)	−0.07 (−0.042)	−0.049*** (−0.044***)
ES	71,995	3.378 (0.397)	4.308 (0.169)	−0.930** (0.228***)	2.993 (0.412)	5.082 (0.348)	−2.089*** (0.064***)
Volume	75,444	4.14×10^7 (7.59×10^5)	1.07×10^8 (1.63×10^7)	$-6.59 \times 10^{7***}$ ($-1.55 \times 10^{7***}$)	3.15×10^7 (1.20×10^6)	6.82×10^7 (5.08×10^6)	$-3.67 \times 10^{7***}$ ($-3.88 \times 10^{6***}$)
Ln(Vol)	75,444	13.732 (13.539)	15.809 (16.608)	−2.077*** (−3.069***)	14.149 (14.00)	15.284 (15.441)	−1.135*** (−1.441***)
Turnover	65,067	0.101 (0.001)	0.204 (0.031)	−0.103*** (−0.030***)	0.068 (0.003)	0.133 (0.008)	−0.064*** (−0.005***)
Fund Return	75,444	0.003 (0.001)	0.006 (0.003)	−0.003*** (−0.002***)	−0.001 (0.000)	−0.002 (−0.001)	0.001*** (0.001)
Stdret	72,756	0.004 (0.002)	0.005 (0.004)	−0.001*** (−0.002***)	0.005 (0.003)	0.006 (0.005)	−0.001*** (−0.002***)
Alpha	75,442	0.0003 (−0.0001)	−0.0003 (−0.001)	0.001* (0.0009***)	0.000 (0.000)	−0.002 (−0.002)	0.002*** (0.002***)
Premium	47,954	0.059 (−0.010)	0.082 (0.072)	−0.023*** (−0.082***)	0.032 (0.001)	0.027 (0.010)	0.005*** (−0.009***)
LD_Before_MktCrash	5,965	3.566 (3.494)	3.041 (3.289)	0.525*** (0.205***)			
Quarterly frequency							
Fund Flow	1,448	−0.039 (−0.045)	0.082 (−0.054)	−0.121** (0.009)	−0.037 (−0.042)	−0.042 (−0.067)	0.005 (0.025**)
Stock Holdings Change	1,030	0.114 (0.006)	0.155 (0.013)	−0.041 (−0.007)	0.002 (−0.012)	0.109 (−0.020)	−0.107*** (0.008)
Cash Holdings Change	1,378	0.019 (0.000)	0.070 (0.008)	−0.051*** (−0.008**)	0.000 (−0.003)	0.004 (−0.007)	−0.004 (0.004***)
Quarterly Fund Return	1,547	0.126 (0.098)	0.214 (0.154)	−0.087*** (−0.056*)	0.024 (0.016)	−0.002 (−0.005)	0.026** (0.021***)
Avg. Alpha	1,355	0.001 (0.001)	0.001 (0.000)	0.000 (0.001*)	0.000 (0.000)	−0.003 (−0.001)	0.003*** (0.001***)
Ret Nav	1,583	0.036 (0.022)	0.050 (0.023)	−0.014** (−0.001)	0.010 (0.009)	−0.169 (−0.033)	0.179*** (0.042***)
Ln(Size)	1,585	20.076 (20.204)	19.706 (20.240)	0.370* (0.040)	19.792 (19.769)	19.932 (19.878)	−0.140 (−0.109)
Ln(Family Size)	1,587	24.428 (24.478)	24.448 (24.634)	−0.020 (−0.156)	25.177 (25.169)	25.385 (25.708)	−0.207*** (−0.539***)
Stockholdings_Before_MktCrash	300	0.370 (0.019)	0.804 (0.900)	−0.434*** (−0.881***)			
Panel B: Time invariant							
		Nondeleveraging			Deleveraging		Difference
Fund Fee	183		0.898 (1.000)			0.981 (1.000)	−0.083*** (0.000)
Fund Age	183		4.443 (4.510)			3.082 (2.392)	1.361*** (2.118***)
Ln(Age)	183		1.406 (1.506)			1.067 (0.872)	0.339*** (0.634***)

Notes. This table reports the descriptive statistics of our sample. The sample consists of 183 structured funds; sample period is from June 1, 2014 to December 31, 2016. Panel A provides the means and medians (in parentheses) of the time-varying variables for the deleveraging funds and the nondeleveraging funds respectively, both before and after June 12, 2015. We also provide the differences in means and medians between the two groups of funds. Panel B provides the means and medians (in parentheses) of the time-invariant variables for the deleveraging funds and the nondeleveraging funds respectively, as well as the differences in means and medians between the two groups of funds. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

trading volume (*Volume* and *Ln(Vol)*), higher turnover (*Turnover*), higher daily fund returns (*Fund Return*),¹⁵ higher return volatility (*Stdret*), lower daily fund alpha (*Alpha*), higher trading premium (*Premium*), higher quarterly fund flow (*Fund Flow*), higher stock holding changes (*Stock Holdings Change*), higher cash holding changes (*Cash Holdings Changes*), higher quarterly fund returns (*Quarterly Fund Return*), higher net profitability (*Ret Nav*), shorter average distance to deleveraging threshold in the 90 trading days before the debut of the market crash (*LD_Before_MktCrash*), and greater average stock holdings in the three quarters before the debut of the market crash (*Stockholdings_Before_MktCrash*).

In the subperiod after June 12, 2015, compared with the nondeleveraging funds, the deleveraging funds on average have lower leverage (*Leverage*), higher order imbalance (*OI*), higher effective spread (*ES*), higher daily trading volume (*Volume* and *Log(Vol)*), higher turnover (*Turnover*), lower daily fund returns (*Fund Return*), higher return volatility (*Stdret*), lower daily fund alpha (*Alpha*), lower trading premium (*Premium*), lower quarterly fund flow (*Fund Flow*), higher stock holding changes (*Stock Holdings Change*), higher cash holding changes (*Cash Holdings Changes*), lower quarterly fund returns (*Quarterly Fund Return*), and lower net profitability (*Ret Nav*).

The deleveraging funds also have higher fund fee (*Fund Fee*) and are younger in fund age (*Fund Age* and *Ln(Age)*) than the nondeleveraging funds.¹⁶ The two groups have similar average fund size (*Ln(Size)*) and family size (*Ln(Family Size)*). As univariate comparisons can mask the true relations, we next conduct rigorous analyses in subsequent sections to thoroughly investigate how deleveraging affects the market liquidity of tranche B equity units and the subsequent funding liquidity of the deleveraged funds.

4. Leverage and Market Liquidity of Fund Equity Units

Leverage-constrained investors' strong preference for high-embedded-leverage securities (Baker et al. 2011;

Frazzini and Pedersen 2012, 2014; Christoffersen and Simutin 2017) can increase the popularity of such securities and hence lead to greater market liquidity for such securities. To show evidence within our empirical setting, we compare structured equity funds versus nonstructured equity funds on their individual ownership over our sample period (2014–2016) in Panel A of Table 2. We find that structured funds attract significantly higher (lower) individual ownership (institutional ownership) than nonstructured funds. The average individual ownership of tranche B equity units of structured equity funds is 87.80%, whereas the average individual ownership of no-structured equity funds is 71.91%. The difference is significant at the 1% level. Moreover, the greater the leverage ratio of a structured equity fund, the higher the individual ownership of the fund's tranche B units, as shown in Panel B of Table 2. As individual investors are typically more leverage-constrained than institutional investors, the finding is consistent with leveraged securities attract investors who are leverage-constrained. Consistent with our expectation, tranche A debt units have very low average individual ownership (35.63% in Panel A) and the individual ownership of debt units also decreases with fund leverage ratio (Panel B).

We then explore the relation between leverage and market liquidity of fund's equity units before the stock market crash and subsequent occurrence of forced deleveraging events. We conjecture a positive relation between the leverage of a structured fund and its equity units' market liquidity. To study this relation, we run the following OLS regression using all fund-day observations in our sample before June 12, 2015.

$$Liquidity_{it} = \alpha + \beta_1 Leverage_{it-1} + \beta_2 Leverage_{it-1}^2 + \gamma X + \varepsilon_{it} \quad (3)$$

In Equation (3), the dependent variable, *Liquidity*, is one of the four daily market-liquidity measures (i.e., *OI*, *ES*, *Ln(Vol)*, or *Turnover*). Our main variable of interest, *Leverage*, is the lagged fund leverage. We expect the regression coefficient β_1 to be significantly positive.

Table 2. Individual Ownership of Structured and Nonstructured Equity Funds

Panel A					
	Nonstructured fund		Tranche A		Tranche B
Individual ownership	71.91%		35.63%		87.80%
Panel B					
	Leverage quintile				
	1	2	3	4	5
Individual ownership: Tranche A	46.21%	35.71%	30.32%	30.78%	26.25%
Individual ownership: Tranche B	86.25%	87.65%	88.04%	88.35%	93.24%

Notes. Panel A of the table reports the average individual ownership of structured equity funds (both tranche A debt units and tranche B equity units) and nonstructured equity funds over the 2014–2016 sample period. Panel B reports the average individual ownership of tranche B equity units of structured equity funds based on leverage quintiles over the sample period.

We further include $Leverage^2$ in the regression to take into account the possibility that a very high level of leverage may reduce market liquidity of tranche B equity units due to the potential concern of crossing the leverage limit and the resulting forced deleveraging. We include a vector of control variables, X , in the regression equation. Following Kacperczyk and Schnabl (2013) and Kahraman and Tookes (2017), these control variables include the lagged measure of market liquidity, lagged fund return (*Fund Return*) and alpha (*Alpha*),¹⁷ the standard deviation of five-minute-interval fund returns during the last trading day (*Stdret*), the natural logarithm of fund AUM at the end of the last quarter ($Ln(Size)$), the natural logarithm of fund family AUM at the end of the last quarter ($Ln(Family\ Size)$), and year-day fixed effects and fund fixed effects. We cluster standard errors at both the fund and year-day levels. Table 3 reports the results from estimating Equation (3).

Columns 2, 4, 6, and 8 show that fund leverage is significantly and positively related to order imbalance, trading volume, and turnover, and significantly and

negatively related to effective spread.¹⁸ Moreover, the coefficient of squared leverage is also statistically significant, with an opposite sign to the coefficient of leverage. This result suggests that investors may be concerned when leverage is very high. However, the absolute value of the squared-leverage coefficient is only around one-fifteenths of the absolute value of the leverage coefficient. This suggests that the inflection point of the leverage–market liquidity relation only occurs when fund leverage is extremely high. For example, our results imply that the relation between leverage and order imbalance only becomes flat or negative when leverage exceeds 8.¹⁹ In that case, leverage already exceeds the limit and triggers forced deleveraging. Moreover, when we exclude the squared leverage term from Equation (3), columns 1, 3, 5, and 7 show that the coefficient of leverage continues to have the same sign as the coefficient when we include the squared term, in the regressions with different dependent variables. However, the coefficient of leverage becomes statistically insignificant for the order imbalance

Table 3. Fund Leverage and Market Liquidity of Equity Units Before the Market Crash

	OI		ES		Ln(Vol)		Turnover	
$Leverage_{it-1}$	0.009 (0.010)	0.065** (0.029)	−0.190 (0.265)	−1.833* (1.097)	0.155* (0.078)	0.780*** (0.133)	0.007* (0.004)	0.047*** (0.012)
$Leverage^2_{it-1}$		−0.004** (0.002)		0.126* (0.074)		−0.047*** (0.009)		−0.003*** (0.001)
OI_{it-1}	0.336*** (0.018)	0.336*** (0.018)						
ES_{it-1}			0.421*** (0.040)	0.421*** (0.040)				
$Ln(Vol)_{it-1}$					0.711*** (0.025)	0.694*** (0.025)		
$Turnover_{it-1}$							0.854*** (0.022)	0.852*** (0.022)
$Fund\ Return_{it-1}$	0.616 (0.399)	0.672* (0.404)	28.47 (19.87)	26.83 (19.61)	0.252 (0.934)	0.945 (0.939)	0.312 (0.190)	0.354* (0.192)
$Stdret_{it-1}$	−4.240*** (1.167)	−4.142*** (1.158)	335.8*** (85.79)	332.9*** (85.83)	−3.638 (3.007)	−1.898 (2.943)	−1.113*** (0.407)	−1.043** (0.403)
$Alpha_{it-1}$	−1.079** (0.467)	−1.128** (0.469)	−38.75* (21.24)	−37.20* (21.07)	−0.502 (1.034)	−1.152 (1.039)	−0.238 (0.198)	−0.276 (0.200)
$Ln(Size)_{it-1}$	0.002 (0.009)	0.0003 (0.009)	−0.550 (0.394)	−0.494 (0.390)	0.113*** (0.034)	0.101*** (0.036)	0.026*** (0.008)	0.025*** (0.008)
$Ln(Family\ Size)_{it-1}$	0.035 (0.036)	0.031 (0.036)	1.114* (0.580)	1.222* (0.574)	0.180 (0.125)	0.149 (0.126)	0.026 (0.024)	0.023 (0.024)
Year-day fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fund fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,503	16,503	16,503	16,503	16,615	16,615	12,924	1,2924
Adjusted R^2	0.182	0.183	0.294	0.294	0.902	0.903	0.900	0.900

Notes. This table reports the results of OLS regressions that investigate the relations between various tranche B market-liquidity measures and fund leverage before the 2015 Chinese stock market crash. In our sample, the regressions use all fund-day observations before June 12, 2015. The dependent variable is *OI*, *ES*, $Ln(Vol)$, or *Turnover*. Our main independent variable of interest is *Leverage*. Control variables include the squared leverage ($Leverage^2$), the lagged measure of market liquidity, lagged fund return (*Fund Return*) and alpha (*Alpha*), the standard deviation of 5-minute-interval fund returns during the last trading day (*Stdret*), the natural logarithm of fund AUM at the end of the last quarter ($Ln(Size)$), the natural logarithm of fund family AUM at the end of the last quarter ($Ln(Family\ Size)$). Year-day fixed effects and fund fixed effects are included but for brevity, their regression coefficients are omitted from reporting. Robust standard errors (in parentheses) are clustered at both the fund and year-day levels. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

and effective spread regressions. This result is understandable because very high levels of leverage may reduce the market liquidity of equity units caused by the potential concern of crossing the leverage limit and the resulting forced deleveraging. Thus, the result is weaker when we use a linear model to fit a potentially nonlinear relation.

Thus, our empirical evidence shows that before exceeding the leverage limit, funds with greater leverage have significantly higher order imbalance, narrower effective spread, greater trading volume, and higher turnover for their equity units. Because these OLS regression results are prone to potential endogeneity issues such as omitted-variable bias, we next use RDD to better identify the leverage effect on the market liquidity of structured funds' equity units.

5. Impact of Deleveraging on the Market Liquidity of Equity Units

We next explore the impact of deleveraging on the market liquidity of a fund's equity units and test Hypothesis 1 (i.e., deleveraging causes a significant deterioration in the market liquidity of the treatment fund's equity units).

5.1. Regression Discontinuity Design

The identification challenge we face is to isolate the impact of deleveraging from the impact of the trading losses on the equity units' market liquidity when structured funds are forced to deleverage due to trading losses. Moreover, other observable/unobservable factors can affect both fund leverage levels and the market liquidity of the funds' equity units, which makes it difficult to establish causality. We tackle these challenges by using RDD to identify the impact of deleveraging on the market liquidity of the equity units.

During the 2015 Chinese stock market crash, some structured funds crossed the deleveraging threshold and were forced to subsequently deleverage. For those funds whose actual leverage is near the deleveraging threshold, it is difficult to precisely control whether the fund's actual leverage crosses the threshold or not. Thus, the treatment can be viewed as being random for such funds near the deleveraging threshold as long as the impact of other observable/unobservable confounding factors on the market liquidity of a fund's equity units is continuous at the threshold. We can then estimate the local average treatment effect by comparing the market liquidity of the equity units of structured funds that are just below the threshold with that of structured funds just crossing the threshold *before the latter funds subsequently deleveraged*.²⁰

There are two ways to implement RDD. The first is to use the full data sample (not just fund-days near the discontinuity point) and higher-order polynomials of the forcing variable in the regressions. The second

is to restrict analysis to the observations around the discontinuity point and estimate the local average treatment effect. The benefit of the first method is that because it uses a large sample, it can potentially reduce noise in estimating the treatment effect. However, including observations far away from the discontinuity point can introduce substantial estimation bias because other factors can play a significant role in influencing the outcome variable. In fact, Gelman and Imbens (2019) show that RDD estimators based on such full-sample methods can be quite misleading; hence they strongly suggest that researchers use the second method. Thus, we focus our analysis on the local sample of fund-day observations near the discontinuity point. Following Kahraman and Tookes (2017), we use the following regression specification with fund-day observations around the discontinuity point to estimate the treatment effect of forced deleveraging on equity units' market liquidity.

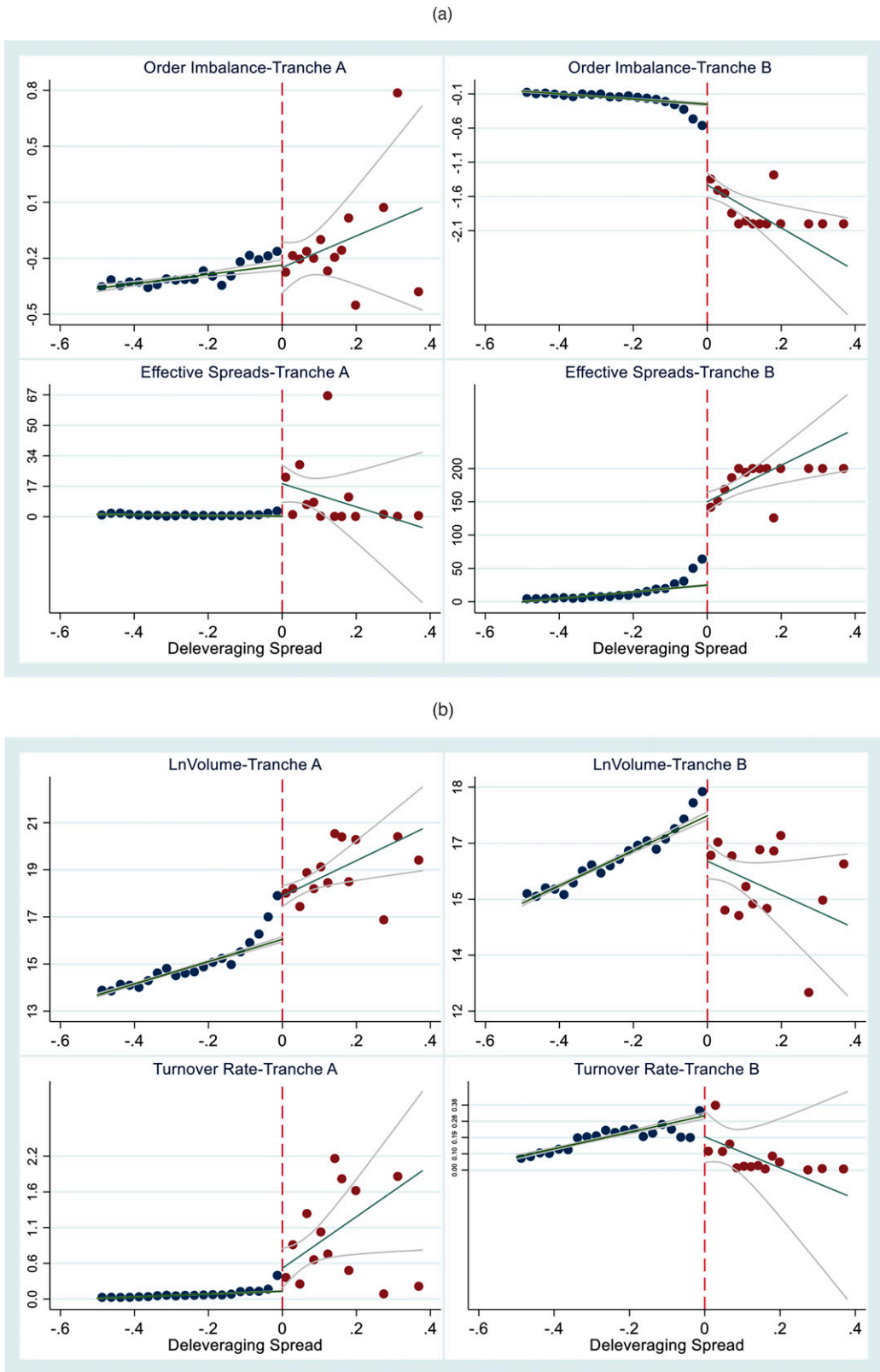
$$Liquidity_{it} = \alpha + \beta Treated_{it} + \gamma'X + \varepsilon_{it} \quad (4)$$

In our RDD framework, the forcing variable is the difference between the specified threshold value of tranche B equity unit and the actual net asset value of tranche B unit (hereafter, deleveraging spread). As mentioned previously (Section 2.1), the fluctuation in actual leverage is mainly driven by changes in the value of tranche B equity unit. Thus, a structured fund typically specifies the threshold net asset value of tranche B equity unit in its prospectus. Once the actual value of tranche B equity unit falls below that threshold, the fund will be forced to recalculate the number of tranche A and B units in the next two trading days and reduce fund leverage to its initial level. If the deleveraging spread is greater than or equal to zero, then deleveraging is triggered and the fund will be forced to deleverage subsequently; that is, $Treated_{it} = 1$. By contrast, if deleveraging spread is less than zero, the fund will not be forced to deleverage and $Treated_{it} = 0$. X is a vector of control variables.

Before estimating Equation (4), it is useful to plot the market liquidity of tranche B equity units around the deleveraging spread threshold. For purpose of comparison, we also plot the market liquidity of tranche A debt units around the threshold. We expect that forced deleveraging will drive away tranche B's risk-loving investors who prefer high embedded leverage, thus leading to a decline in the market liquidity of equity units. However, forced deleveraging should not drive away tranche A's risk-averse fixed-income investors. In fact, deleveraging can make tranche A debt units more attractive to such investors because debt units become safer after deleveraging.

Figure 4 plots the average market liquidity around the deleveraging spread threshold. We examine all fund-days with deleveraging spread between -0.50

Figure 4. (Color online) Average Market Liquidity Around the Deleveraging Spread Threshold



Notes. This figure plots the average market liquidity around the deleveraging spread threshold. We examine all fund-days with deleveraging spread between -0.50 and 0.40 . We form 20 bins on each side of the threshold (i.e., deleveraging spread = 0) and compute the average market liquidity within each bin. We also run separate regressions of average market liquidity on deleveraging spread and plot the regression lines and 95% confidence intervals. (a) Average order imbalance (OI) and effective spreads (ES) around the threshold. (b) Average trading volume ($\text{Ln}(\text{Vol})$) and turnover (Turnover) around the threshold. In each figure, the left panels plot the average market liquidity for tranche A debt units, while the right panels plot the average market liquidity for tranche B equity units. Appendix Table A1 presents detailed description of the variables.

and 0.40. We form 20 bins on each side of the threshold (i.e., deleveraging spread = 0) and compute the average market liquidity within each bin.²¹ We also run separate regressions of average market liquidity on deleveraging spread and plot the regression lines and 95% confidence intervals. Figure 4(a) clearly shows discontinuous drop in order imbalance and discontinuous jump in effective spread for tranche B equity units at the deleveraging spread threshold. This is consistent with a loss of investors' interests and demand of tranche B equity units when investors learn that the fund will subsequently deleverage. By contrast, we do not observe discontinuous changes in order imbalance or effective spread for tranche A debt units at the threshold.

Figure 4(b) shows that trading volume and turnover for tranche B equity units increase with deleveraging spread to the left of the discontinuity point, and there is a drop in tranche B's trading activity to the right of the discontinuity point, with the pattern for trading volume being more obvious than that for turnover. This suggests that trading becomes more active when fund leverage gets closer to the deleveraging spread threshold (likely because higher leverage provides greater benefits to leverage-constrained investors who seek high expected returns). After investors learn that the fund will deleverage subsequently, their interests in tranche B equity units decrease and thus trading activity in these securities decline. In contrast, we do not observe discontinuity in trading volume or turnover for tranche A debt units at the threshold. It seems trading activity for tranche A debt units continues to increase to the right of the threshold. This is likely due to two reasons. First, because tranche A debt units have now become safer given deleveraging, they may also attract more risk-averse investors. Second, deleveraging makes tranche B equity units much less appealing to tranche B investors. Given the significant deterioration of tranche B market liquidity, rather than selling their tranche B units to the market, some investors may choose to purchase tranche A units from the market to combine with their tranche B units into parent units and then redeem their investment from the fund.

Finally, we check for evidence of strategic manipulation of fund leverage near the threshold. Such manipulation may be conducted to avoid forced deleveraging. If such strategic manipulation exists, we should observe a significant bunching in the number of funds immediately below the deleveraging spread threshold (deleveraging spread = 0) compared with the number of funds immediately above the threshold. As shown in Figure 5, we do not observe such obvious bunching. This is not too surprising. Given the trading losses, it would be very difficult, if not impossible, for

fund managers to strategically push fund leverage below the preset limit. This finding further supports the validity of the RDD identification strategy.

5.2. RDD Estimation Results

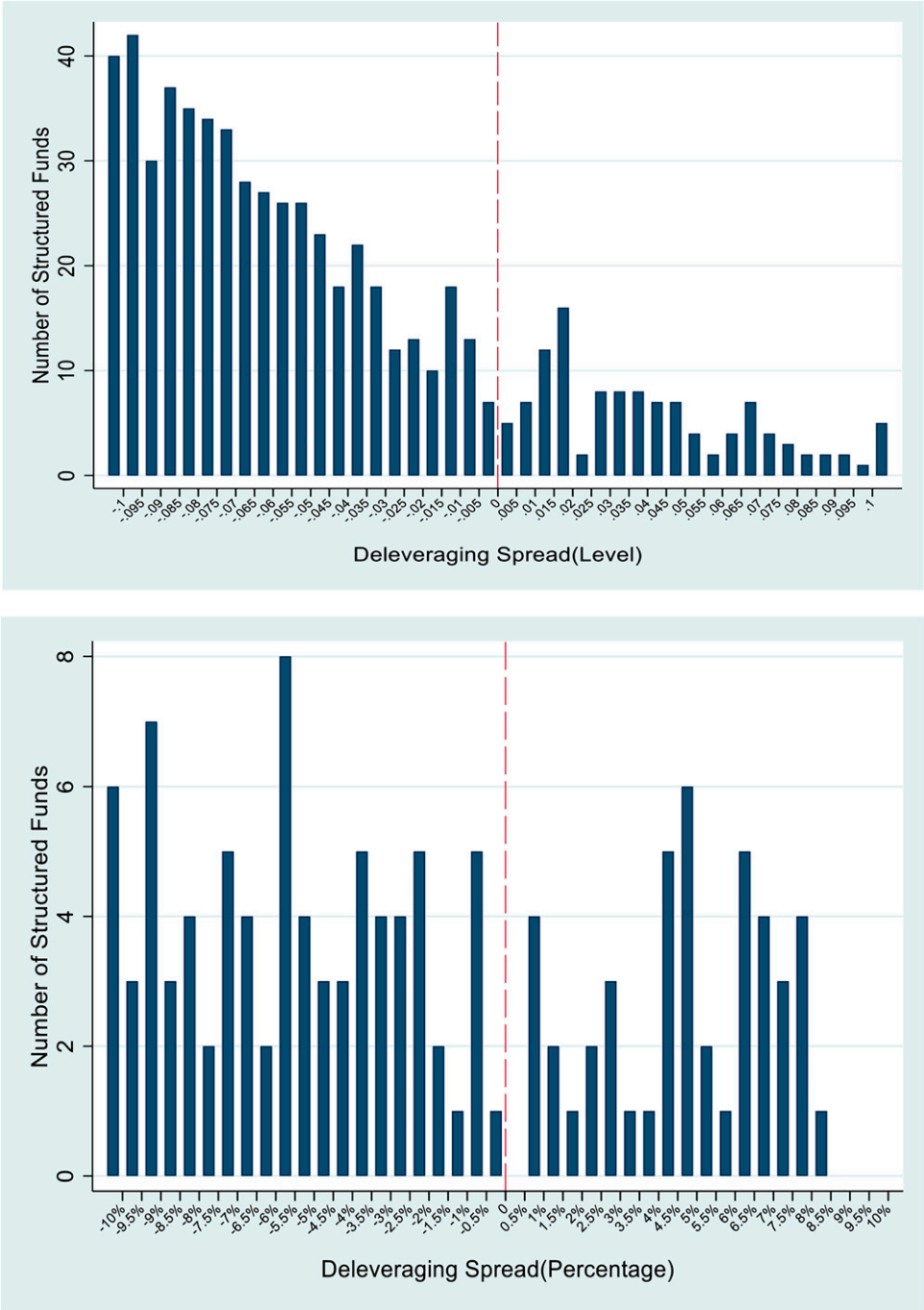
Our RDD analysis uses the sample of fund-day observations around the deleveraging spread threshold. An important issue is the selection of bandwidth. The appropriate bandwidth should be small enough so that the treatment is locally random but large enough so that a sufficient number of observations exist to provide statistical power. Imbens and Kalyanaraman (2012) suggest using the mean squared error (MSE) loss criterion to select an optimal bandwidth that minimizes MSE depending on the initial bandwidth choice. Calonico et al. (2014) improve on Imbens and Kalyanaraman (2012) by providing the optimal choice for initial bandwidth. Calonico et al. (2017) further provide data-driven bandwidth selection for either one common bandwidth or two distinct bandwidths on the two sides of the threshold. Cattaneo et al. (2016) use finite-sample methods to select a narrow window near the threshold where the local randomization assumption is most plausible.

We mainly rely on the data-driven bandwidth selection approach of Calonico et al. (2017) and select one common bandwidth on both sides of the threshold. Because funds that cross the deleveraging spread threshold will be forced to deleverage in the next two trading days, there are much fewer observations on the right-hand side of the threshold (i.e., deleveraging spread > 0) than on the left-hand side of the threshold. For robustness, we also follow Calonico et al. (2017) and select two distinct bandwidths on either side of the threshold to reestimate our RDD regressions.

As our RDD analysis focuses only on fund-day observations near the discontinuity point, treatment can be viewed as random; thus, it is not necessary to include control variables (Lee and Lemieux 2010). In fact, adding controls may introduce biases in estimating the treatment effect if the control variables are affected by the treatment. Nevertheless, adding controls may reduce standard errors and improve the precision in estimating the treatment effect. We thus estimate Equation (4) both with and without controls. Standard errors are clustered at both the fund and year-day levels. Table 4 reports the RDD regression results.²²

Columns 1, 4, and 7 of Table 4 provide the RDD estimation results for different tranche B market-liquidity measures, using one common bandwidth on both sides of the threshold and without controls. The results clearly show a large and highly significant impact of deleveraging on order imbalance, effective spread, and trading volume. Compared with the

Figure 5. (Color online) Fund Distribution Around the Deleveraging Spread Threshold



Notes. This figure plots two histograms of the number of funds near the deleveraging spread threshold to examine whether strategic manipulation of fund leverage exists near the threshold to avoid forced deleveraging. The upper (lower) histogram uses the level distance (the percentage distance) to the threshold.

fund-day observations just below the threshold, when a fund just crosses the deleveraging spread threshold, its order imbalance decreases on average by 2.1 times ($=1.110/0.529$), effective spread increases by 5.8 times ($=111.2/19.031$), and trading volume decreases by

43% ($=1.284/3.023$), relative to their respective standard deviations.

To improve the estimation precision of the treatment effect, in columns 2, 5, and 8, we include lagged fund alpha (*Alpha*), lagged natural logarithm of fund

Table 4. Impact of Deleveraging on Tranche B Market Liquidity

	OI			ES			Ln(Vol)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$Treated_{it}$	−1.110*** (0.130)	−0.658*** (0.207)	−1.031*** (0.157)	111.2*** (13.15)	54.51*** (17.07)	82.84*** (15.70)	−1.284*** (0.438)	−1.035*** (0.220)	−0.926*** (0.260)
OI_{it-1}		0.482*** (0.113)	0.385*** (0.057)						
ES_{it-1}					0.628*** (0.122)	0.619*** (0.096)			
$Ln(Vol)_{it-1}$								0.790*** (0.103)	0.854*** (0.034)
$Alpha_{it-1}$		−0.406 (1.393)	0.257 (0.566)		−119.1 (182.8)	−93.47 (81.71)		0.540 (1.452)	−0.446 (0.994)
$Ln(Size)_{it-1}$		−0.022 (0.035)	0.014** (0.007)		1.855 (2.866)	0.796 (0.796)		0.168 (0.111)	0.172*** (0.046)
$Ln(Family\ Size)_{it-1}$		−0.028 (0.097)	0.001 (0.011)		6.426 (7.901)	0.960 (2.065)		0.058 (0.081)	0.029 (0.034)
$LD_Before_MktCrash_i$		−0.258 (0.237)	0.028* (0.015)		33.83 (22.28)	14.45* (8.511)		0.435 (0.349)	0.105* (0.060)
$Stockholdings_Before_MktCrash_i$		0.901 (1.239)	−0.091 (0.064)		−130.0 (107.5)	−66.19 (40.11)		−1.334 (1.393)	−0.079 (0.189)
$Fund\ Fee_i$		0.416 (0.732)	−0.124** (0.055)		−34.56 (79.92)	1.985 (4.404)		−0.013 (0.186)	−0.042 (0.082)
$Ln(Age)_i$		0.431* (0.233)	0.034 (0.040)		−37.15* (21.41)	−2.716 (6.484)		0.262* (0.142)	−0.032 (0.093)
Bandwidth	[−0.077, 0.077]	[−0.077, 0.077]	[−0.312, 0.081]	[−0.071, 0.071]	[−0.071, 0.071]	[−0.224, 0.081]	[−0.114, 0.114]	[−0.114, 0.114]	[−0.276, 0.094]
Observations	658	339	3,997	555	283	2,071	1,367	615	3,231
Adjusted R^2	0.218	0.409	0.207	0.269	0.577	0.462	0.026	0.869	0.934

Notes. This table reports the results of RDD regressions that investigate the impact of deleveraging on tranche B's market liquidity. The RDD regression sample includes all fund-day observations with deleveraging spread (i.e., the forcing variable) close to the threshold of zero. The dependent variable is OI , ES , or $Ln(Vol)$. The main independent variable of interest is $Treated$, which equals one if deleveraging spread of a fund-day is greater than the threshold value of zero (and hence the fund needs to subsequently deleverage) and equals zero otherwise. Control variables include the lagged dependent variable, lagged fund alpha ($Alpha$), lagged natural logarithm of fund AUM ($Ln(Size)$), lagged natural logarithm of fund family AUM ($Ln(Family\ Size)$), average distance of the fund's actual leverage to its deleveraging threshold in the 90 trading days immediately before June 12, 2015 ($LD_Before_MktCrash$), the fund's average total value of quarter-end stock holdings scaled by total assets under management in the three quarters immediately before June 12, 2015 ($Stockholdings_Before_MktCrash$), fund management fee ($Fund\ Fee$), and natural logarithm of fund age ($Ln(Age)$). We follow the data-driven bandwidth selection approach of Calonico et al. (2017) and select one common bandwidth on both sides of the threshold or two distinct bandwidths on the two sides of the threshold. The selection of bandwidth is reported in brackets. Robust standard errors (in parentheses) are clustered at both the fund and year-day levels. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

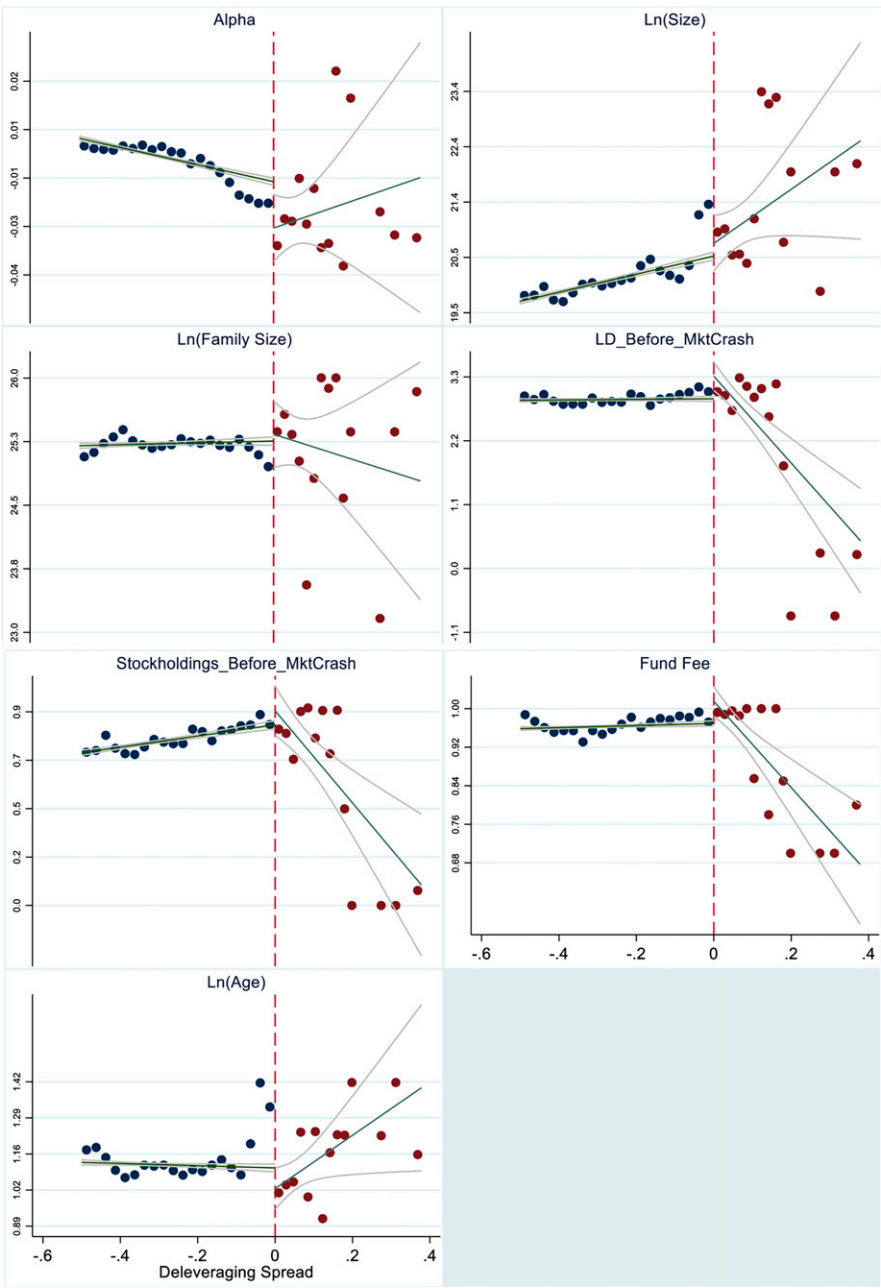
AUM ($Ln(Size)$), lagged natural logarithm of fund family AUM ($Ln(Family\ Size)$), average distance of the fund's actual leverage to its deleveraging threshold in the 90 trading days immediately before June 12, 2015 ($LD_Before_MktCrash$), the fund's average total value of quarter-end stock holdings scaled by total assets under management in the three quarters immediately before June 12, 2015 ($Stockholdings_Before_MktCrash$), natural logarithm of fund age ($Ln(Age)$), fund management fee ($Fund\ Fee$), and the lagged dependent variable as control variables. Figure 6 shows plots for these control variables for fund-day observations around the discontinuity point.

We do not observe discontinuous change for any of these control variables, which suggests that adding these controls will not introduce biases in the

estimation of treatment effect. The RDD regression results with controls are qualitatively similar to those without controls.²³ In columns 3, 6, and 9, we reestimate the RDD regressions using two distinct bandwidths on the two sides of the threshold. Although the number of fund-day observations significantly increases in these regressions, the results are qualitatively similar to those based on using one common bandwidth on both sides of the threshold, indicating that the finding is insensitive to alternative bandwidths.

As high leverage attracts individual investors, some may argue that an alternative channel may potentially explain the findings. Individuals are subject to ambiguity aversion (Ellsberg 1961) and other behavioral biases that can cause a sudden drop in demand following losses. Following losses, deleveraging happened while,

Figure 6. (Color online) Control Variables of the RDD Regressions Around the Deleveraging Spread Threshold



Notes. This figure plots the control variables of our RDD regressions for fund-day observations around the deleveraging spread threshold. We examine all fund-days with deleveraging spread between -0.50 and 0.40 . We form 20 bins on each side of the threshold (i.e., deleveraging spread = 0) and compute the average value of the control variable within each bin. We also run separate regressions of the average value of the control variable on deleveraging spread and plot the regression lines and 95% confidence intervals. Table A.1 presents detailed description of the variables.

at the same time, individual investors panicked, resulting in market liquidity reduction and large withdrawal. However, this channel can be ruled out by the RDD results which show the discontinuity between deleveraging funds and nondeleveraging funds (as both groups have similar performance and other characteristics).²⁴

Classic theories on trading volume (Kyle 1985, Admati and Pfleiderer 1988) suggest that greater volume is related to higher liquidity because trading volume is mostly driven by liquidity traders. For example, Admati and Pfleiderer (1988) suggest that liquidity traders tend to trade together, which leads to volume concentration and

an increase in market liquidity. For trading volume, indeed we observe clear discontinuity (i.e., obvious drop) once the fund crosses the deleveraging threshold (as shown in the top-right graph of Figure 4(b)). Similarly, when trading volume is the dependent variable, the regression coefficient of the *Treated* indicator is negative and highly statistically significant at the 1% level in columns 7–9 of Table 4. However, the result on turnover is weak (as shown in the bottom-right graph of Figure 4(b)). The drop in turnover is weaker than the drop in trading volume (when the fund crosses the deleveraging threshold), likely because daily turnover is daily trading volume scaled by the total number of tranche B equity units outstanding. In our setting, the daily total number of tranche B units outstanding can change. As shown in Tables 8–11, after the fund crosses the deleveraging threshold, investors increase their redemption demand (i.e., fund outflows), which reduces the total number of tranche B units outstanding. Thus, the drop in turnover is smaller since the drop in the numerator of turnover is partially offset by the drop in the denominator.

To further ensure the robustness of our finding, we follow the approach of Cattaneo et al. (2016) for selecting a narrow window near the threshold where the local randomization assumption is most plausible, and we use their local randomization inference method to estimate the local average treatment effect. The results provided in Table A.3 in the Appendix are similar to those based on OLS regressions in Table 4. Within this narrow window around the threshold, the local average treatment effect is also statistically significant at the 5% level for turnover.

As the vast majority of structured funds set the leverage limit to five, we further exclude all structured funds with leverage limits not being five from the sample; the RDD regression results shown in Table A.4 in the Appendix are qualitative very similar to those in Table 4.

5.3. Alternative Managerial-Quality-Signaling Explanation

The RDD estimation results suggest that when a fund crosses the deleveraging spread threshold, the fund's equity units lose their attractiveness to investors who are seeking high expected returns (because of the expected subsequent deleveraging), and this results in a significant deterioration in the market liquidity of the fund's equity units. However, an alternative managerial-quality-signaling explanation may explain the results. Specifically, a fund crossing the leverage limit may signal to investors that the fund's managerial quality is low. Managerial quality signaling, rather than expected subsequent deleveraging, can lead to a

deterioration of the market liquidity of the fund's equity units. We implement two tests to investigate this alternative explanation.

First, we look at the fund's performance uncertainty before the market crash. We rank the sample funds according to their Tranche B performance (alpha) each day in the 90 days immediately before June 12, 2015—greater rank indicates better performance. We classify funds into high and low performance uncertainty and construct a *High Performance Uncertainty* indicator (which equals one if the volatility of the fund's daily performance rank in these 90 days is greater than the sample median and equals zero otherwise) to interact with the *Treatment* indicator. The managerial quality signal to investors should be stronger when investors have greater uncertainty on a fund's managerial quality. Thus, we would expect the treatment effect to be stronger in high performance uncertainty fund if managerial ability signaling is the main driving force of the results.

Second, we look at the fund's past performance before the market crash. We classify funds into high and low past performance and construct a *High Past Performance* indicator (which equals one if the fund's average daily performance rank in the 90 days before June 12, 2015 is greater than the sample median and equals zero otherwise) to interact with *Treatment*. As crossing the deleveraging threshold may constitute a surprise for funds that have strong past performance, we would expect that the managerial ability signal, and thus the treatment effect, may be stronger for such funds than for weak performance funds if the signaling story is true. The results are reported in Table 5.²⁵

Panel A of Table 5 shows that the coefficient of *High Performance Uncertainty* $\times$ *Treatment* is significantly positive (significantly negative) when the dependent variable is order imbalance or trading volume (effective spread), indicating that the treatment effect on market liquidity is weaker for funds with greater performance uncertainty, inconsistent with the managerial ability signaling explanation. Moreover, Panel B shows that the coefficient of *High Past Performance* $\times$ *Treatment* is insignificant for all regressions, again inconsistent with the alternative signaling explanation.²⁶

5.4. Alternative Trading-Premium Explanation

Tranche B units are on average traded with premiums in the market (likely because of their high embedded leverage). Deleveraging will result in a loss of such premiums (Gao et al. 2020). It is likely that the anticipated trading premium loss, not deleveraging itself, results in the reduction in tranche B market liquidity after the fund crosses the threshold. If this alternative

Table 5. Testing the Alternative Managerial-Quality-Signaling Explanation

	<i>OI</i>	<i>ES</i>	<i>Ln(Vol)</i>
Panel A			
<i>Treated_{it}</i>	−1.119*** (0.201)	102.2*** (14.29)	−1.436*** (0.203)
<i>High Performance Uncertainty_i</i>	−0.463** (0.172)	49.66*** (11.23)	−0.401** (0.186)
<i>Treated_{it} × High Performance Uncertainty_i</i>	0.710*** (0.261)	−75.30*** (19.29)	0.539* (0.279)
Controls	Yes	Yes	Yes
Bandwidth	[−0.077, 0.077]	[−0.071, 0.071]	[−0.114, 0.114]
Observations	339	283	615
Adjusted <i>R</i> ²	0.435	0.613	0.871
Panel B			
<i>Treated_{it}</i>	−0.600*** (0.199)	50.06*** (14.92)	−0.941*** (0.293)
<i>High Past Performance_i</i>	−0.343** (0.150)	35.40*** (7.321)	−0.047 (0.130)
<i>Treated_{it} × High Past Performance_i</i>	−0.0337 (0.213)	0.474 (13.64)	−0.239 (0.357)
Controls	Yes	Yes	Yes
Bandwidth	[−0.077, 0.077]	[−0.071, 0.071]	[−0.114, 0.114]
Observations	339	283	615
Adjusted <i>R</i> ²	0.429	0.599	0.869

Notes. This table reports the results of RDD regressions that investigate the impact of deleveraging on tranche B's market liquidity. The RDD regression sample includes all fund-day observations with deleveraging spread (i.e., the forcing variable) close to the threshold of zero. The dependent variable is *OI*, *ES*, or *Ln(Vol)*. The main independent variable of interest is *Treated* × *High Performance Uncertainty* in Panel A and *Treated* × *High Past Performance* in Panel B, respectively. *Treated* is defined as in Table 4. We rank the sample funds according to their tranche B performance (alpha) each day in the 90 days immediately before June 12, 2015—greater rank indicates better performance. *High Performance Uncertainty* equals one if the volatility of the fund's daily performance ranking in these 90 days is greater than the sample median and equals zero otherwise. *High Past Performance* equals one if the fund's average daily performance ranking in these 90 days is greater than the sample median and equals zero otherwise. We follow the data-driven bandwidth selection approach of Calonico et al. (2017) and select one common bandwidth on both sides of the threshold. Robust standard errors (in parentheses) are clustered at both the fund and year-day levels. Control variables (as in Table 4) are included, but for brevity, their regression coefficients are omitted from reporting. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

trading-premium explanation is true, we expect the reduction in tranche B market liquidity (after the fund crosses the deleveraging threshold) to be greater for funds with higher ex ante tranche B trading premiums. We thus construct an indicator variable, *High Prem*, which equals 1 if a fund's average tranche B premium (i.e., (price per unit – net asset value per unit)/net asset value per unit) over the past five trading days is greater than the median average premium of all fund-day observations in the left side of the deleveraging spread threshold in the respective RDD regression sample and equals 0 otherwise.²⁷ We then interact *High Prem* with *Treated* in the RDD regressions. If the alternative explanation is true, we would expect to see the coefficient of *Treated* × *High Prem* to be significantly negative (significantly positive) for the *OI* and *Ln(Vol)* regressions (for the *ES* regression). However, Table A.5 in the Appendix show that the coefficient of *Treated* × *High Prem* is insignificantly positive for the *OI* and *Ln(Vol)* regressions and significantly

negative for the *ES* regression, contradicting the alternative trading-premium explanation.

5.5. Alternative Arbitrage-Trading Explanation

Another alternative explanation is that investor arbitrage trading behavior may drive our findings. Specifically, arbitrageurs, who take positions of tranche M parent units, can form expectations about the potential trigger of the forced deleveraging (when the net asset value of fund equity unit falls below the deleveraging threshold). When these arbitrageurs expect the price of tranche B equity unit to drop below the deleveraging threshold and thus deleveraging to occur in the near future, they quickly split the parent units into A and B units and sell them in the market. Their arbitrage trading behavior can then negatively impact market liquidity. That is, the documented market liquidity deterioration of tranche B equity units may be caused by investor arbitrage trading behavior rather than by deleveraging.

We argue that our finding of a significant treatment effect of deleveraging on tranche B market liquidity is unlikely to be driven by such arbitrage trading behavior. Our RDD methodology examines the average treatment effect of deleveraging within a very small neighborhood around the deleveraging threshold. The ex ante likelihood that deleveraging will occur in the future, and thus the intensity of the arbitrage trading activity should be very similar for funds within this very narrow neighborhood. The RDD results are hence consistent with a causal interpretation that deleveraging negatively impacts the market liquidity of structured funds' equity units because of investors' preferences for high-embedded-leverage securities.

Moreover, the arbitrage trading explanation suggests that when the arbitrageurs expect deleveraging to

occur in the near future, they quickly split the parent units into A and B units and sell them in the market. Hence, we should also expect a negative impact of deleveraging on tranche A debt units of the treatment funds. We thus conduct placebo RDD tests to examine the impact of forced deleveraging on the market liquidity of treatment funds' tranche A debt units. The results are reported in Table 6.

We do not find any significant impact of deleveraging on the order imbalance and effective spread of treatment funds' debt units. By contrast, we find a positive impact of deleveraging on trading volume of the treatment funds' debt units, which is also consistent with the patterns shown in Figure 4. These findings suggest that the arbitrage trading explanation is unlikely to drive our results.²⁸

Table 6. Placebo RDD Test on the Impact of Deleveraging on Tranche A Market Liquidity

	OI			ES			Ln(Vol)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Treated_{it}</i>	−0.024 (0.127)	0.025 (0.117)	0.061 (0.130)	14.66 (11.49)	15.90 (10.97)	18.11 (13.82)	1.265** (0.588)	0.761*** (0.151)	1.144*** (0.175)
<i>OI_{it−1}</i>		0.374*** (0.048)	0.531*** (0.037)						
<i>ES_{it−1}</i>					0.114 (0.273)	0.306*** (0.093)			
<i>Ln(Vol)_{it−1}</i>								0.686*** (0.113)	0.755*** (0.044)
<i>Alpha_{it−1}</i>		−6.632* (3.582)	−6.855*** (1.822)		−138.2 (294.6)	−27.03 (57.38)		2.870 (3.475)	1.243 (3.550)
<i>Ln(Size)_{it−1}</i>		−0.014 (0.017)	−0.0004 (0.012)		1.599 (1.393)	−0.339** (0.151)		0.342*** (0.121)	0.285*** (0.066)
<i>Ln(Family Size)_{it−1}</i>		0.003 (0.050)	−0.018 (0.029)		−1.195 (2.854)	0.133 (0.249)		−0.079 (0.076)	0.082 (0.061)
<i>LD_Before_MktCrash_i</i>		0.440** (0.184)	0.127 (0.086)		−1.281 (8.160)	−0.008 (0.138)		1.213*** (0.439)	0.268* (0.155)
<i>Stockholdings_Before_MktCrash_i</i>		−1.815*** (0.658)	−0.288 (0.355)		11.61 (36.43)	0.515 (0.414)		−4.331*** (1.547)	−0.996 (0.643)
<i>Fund Fee_i</i>		0.025 (0.145)	−0.010 (0.029)		−2.373 (9.453)	0.348 (0.486)		−0.318 (0.698)	−0.173 (0.168)
<i>Ln(Age)_i</i>		0.242** (0.105)	0.008 (0.079)		−12.07 (8.382)	−0.071 (0.379)		−0.005 (0.175)	0.120 (0.152)
Bandwidth	[−0.074, 0.074]	[−0.074, 0.074]	[−0.272, 0.076]	[−0.075, 0.075]	[−0.075, 0.075]	[−0.575, 0.075]	[−0.073, 0.073]	[−0.073, 0.073]	[−0.264, 0.071]
Observations	606	309	2953	630	323	10217	606	301	2866
Adjusted <i>R</i> ²	−0.001	0.173	0.331	0.055	0.073	0.129	0.041	0.879	0.847

Notes. This table reports the results of RDD regressions that investigate the impact of deleveraging on tranche A's market liquidity. The RDD regression sample includes all fund-day observations with deleveraging spread (i.e., the forcing variable) close to the threshold of zero. The dependent variable is *OI*, *ES*, or *Ln(Vol)*. The main independent variable of interest is *Treated*, which equals one if deleveraging spread of a fund-day is greater than the threshold value of zero (and hence the fund needs to subsequently deleverage) and equals zero otherwise. Control variables include the lagged dependent variable, lagged fund alpha (*Alpha*), lagged natural logarithm of fund AUM (*Ln(Size)*), lagged natural logarithm of fund family AUM (*Ln(Family Size)*), average distance of the fund's actual leverage to its deleveraging threshold in the 90 trading days immediately before June 12, 2015 (*LD_Before_MktCrash*), the fund's average total value of quarter-end stock holdings scaled by total assets under management in the three quarters immediately before June 12, 2015 (*Stockholdings_Before_MktCrash*), fund management fee (*Fund Fee*), and natural logarithm of fund age (*Ln(Age)*). We follow the data-driven bandwidth selection approach of Calonico et al. (2017) and select one common bandwidth on both sides of the threshold or two distinct bandwidths on the two sides of the threshold. The selection of bandwidth is reported in brackets. Robust standard errors (in parentheses) are clustered at both the fund and year-day levels. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

5.6. Alternative Adverse-Selection Explanation

The evidence on order imbalance and effective spread of tranche B equity units suggests that market makers are asking for a larger compensation for their inventory holdings when the demand on the tranche B units declines. There can be an alternative explanation that is not based on leverage-constrained investors' demand. Specifically, the forced deleveraging, as a salient event, can trigger the attention of market makers, who become suddenly concerned about the quality of the underlying assets of these leveraged products. As a result, they charge larger ask-bid spreads in fear of adverse selection.²⁹

To investigate this alternative explanation, we use tick-by-tick trading data of the tranche B equity units of the structured funds in our sample and decompose the daily effective spread into three proportions—the inventory component, the order processing component, and the adverse selection component following Huang and Stoll (1997). Specifically, following Huang and Stoll (1997), we first bunch sequential trades as one trade when the transactions have the same bid-ask spread, execution price, and trade direction. We then estimate the inventory component, the order processing component, and the adverse selection component from the daily trades of the funds' tranche B equity units by jointly estimating

Table 7. Impact of Deleveraging on Tranche B Effective Spread Components

	Inventory Holding		Adverse Selection		Order Processing	
<i>Treated_{it}</i>	0.329*** (0.118)	0.311** (0.125)	0.009 (0.058)	−0.098 (0.078)	−0.308*** (0.090)	−0.323*** (0.110)
<i>Inventory Holding_{it−1}</i>		0.027 (0.055)				
<i>Adverse Selection_{it−1}</i>				−0.002 (0.042)		
<i>Order Processing_{it−1}</i>						0.185 (0.119)
<i>Alpha_{it−1}</i>		−1.288 (0.867)		−0.810 (0.777)		−0.321 (1.177)
<i>Ln(Size)_{it−1}</i>		−0.005 (0.032)		−0.045*** (0.012)		0.019 (0.023)
<i>Ln(Family Size)_{it−1}</i>		−0.163 (0.100)		0.096** (0.041)		−0.001 (0.073)
<i>LD_Before_MktCrash_{it}</i>		0.089 (0.217)		0.087 (0.148)		0.104 (0.196)
<i>Stock_Before_MktCrash_{it}</i>		−1.148 (0.927)		−0.433 (0.660)		−0.292 (0.988)
<i>Fund Fee_{it}</i>		0.321* (0.182)		−0.213 (0.233)		−0.710*** (0.200)
<i>Ln(Age)_{it}</i>		−0.503 (0.298)		0.217* (0.115)		0.196 (0.245)
Bandwidth	[−0.071, 0.071]					
Observations	249	141	284	160	346	180
Adjusted R ²	0.047	0.512	0.000	0.215	0.060	0.335

Notes. This table reports the results of RDD regressions that investigate the impact of deleveraging on tranche B's effective spread components. We use tick-by-tick trading data of the tranche B equity units of the structured funds in our sample and decompose the daily effective spread into three proportions—the inventory component, the order processing component, and the adverse selection component following Huang and Stoll (1997). Specifically, following Huang and Stoll (1997), we first bunch sequential trades as one trade when the transactions have the same bid-ask spread, execution price, and trade direction. We then estimate the inventory component, the order processing component, and the adverse selection component from the daily trades of the funds' tranche B equity units by jointly estimating equations (26) and (21) of Huang and Stoll (1997) using seemingly unrelated regressions. We require the estimated component values to be nonnegative by dropping negative values. We then separately examine the impact of deleveraging on each effective spread component. The RDD regression sample includes all fund-day observations with deleveraging spread (i.e., the forcing variable) close to the threshold of zero. The dependent variable is one of the estimated effective spread components (i.e., inventory holding, adverse selection, or order processing). The main independent variable of interest is *Treated*, which equals one if deleveraging spread of a fund-day is greater than the threshold value of zero (and hence the fund needs to subsequently deleverage) and equals zero otherwise. Control variables include the lagged dependent variable, lagged fund alpha (*Alpha*), lagged natural logarithm of fund AUM (*Ln(Size)*), lagged natural logarithm of fund family AUM (*Ln(Family Size)*), average distance of the fund's actual leverage to its deleveraging threshold in the 90 trading days immediately before June 12, 2015 (*LD_Before_MktCrash*), the fund's average total value of quarter-end stock holdings scaled by total assets under management in the three quarters immediately before June 12, 2015 (*Stockholdings_Before_MktCrash*), fund management fee (*Fund Fee*), and natural logarithm of fund age (*Ln(Age)*). We follow Calonico et al.'s (2017) data-driven bandwidth selection approach and select one common bandwidth on both sides of the threshold. The selection of bandwidth is reported in brackets. Robust standard errors (in parentheses) are clustered at both the fund and year-day levels. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 8. Impact of Deleveraging on Funding Liquidity

	Fund Flow	Stock Holdings Change	Cash Holdings Change	Quarterly Fund Return
$Treated_{ic} \times Post_{ct}$	-0.343*** (0.092)	-0.358*** (0.108)	-0.104*** (0.023)	-0.397*** (0.067)
$Post_{ct}$	-0.068* (0.037)	-0.016 (0.050)	-0.017 (0.011)	-0.006 (0.024)
$Treated_{ic}$	0.255*** (0.087)	0.306*** (0.093)	0.076*** (0.022)	0.168*** (0.046)
$Avg. Alpha_{ict-1}$	4.141 (3.868)	-3.954 (5.102)	-1.241 (0.982)	-17.94*** (5.223)
$Ret Nav_{ict-1}$	-0.183 (0.136)	-0.815*** (0.113)	-0.003 (0.029)	-0.281*** (0.077)
$Ln(Size)_{ict-1}$	-0.021* (0.011)	-0.022** (0.010)	-0.011*** (0.003)	0.002 (0.005)
$Ln(Family Size)_{ict-1}$	0.011 (0.010)	-0.009 (0.015)	0.004 (0.003)	0.008 (0.007)
$LD_Before_MktCrash_{ic}$	-0.007 (0.008)	-0.010 (0.014)	-0.001 (0.003)	-0.003 (0.005)
$Stockholdings_Before_MktCrash_{ic}$	-0.064 (0.053)	-0.140** (0.069)	0.002 (0.013)	0.140*** (0.025)
$Fund Fee_{ic}$	0.162* (0.084)	0.182** (0.089)	0.013 (0.016)	-0.049 (0.041)
$Ln(Age)_{ic}$	-0.210*** (0.062)	-0.220*** (0.080)	-0.042*** (0.016)	-0.034 (0.030)
Cohort fixed effects	Yes	Yes	Yes	Yes
Year-Qtr fixed effects	Yes	Yes	Yes	Yes
Observations	1,030	741	1,046	1,142
Adjusted R^2	0.113	0.365	0.124	0.491

Notes. This table reports the results of DID regressions that investigate the impact of deleveraging on fund flows, stock holdings changes, cash holdings changes, and fund returns in subsequent quarters. We use three quarters before and three quarters after the deleveraging-event quarter (the deleveraging-event quarter is excluded) to construct cohorts, and then stack all cohorts into a sample. The dependent variable is *Fund Flow*, *Stock Holdings Change*, *Cash Holdings Change*, or *Quarterly Fund Return*. Our main independent variable of interest is $Treated \times Post$. *Treated* equals one if fund i in cohort c is the treatment fund and equals zero otherwise. *Post* equals one if quarter t is a quarter after the deleveraging event in cohort c and equals zero otherwise. Control variables include the lagged average fund alpha ($Avg. Alpha$), lagged return on net asset value ($Ret Nav$), lagged natural logarithm of fund AUM ($Ln(Size)$), lagged natural logarithm of fund family AUM ($Ln(Family Size)$), average distance of the fund's actual leverage to its deleveraging threshold in the 90 trading days immediately before June 12, 2015 ($LD_Before_MktCrash$), the fund's average total value of quarter-end stock holdings scaled by total assets under management in the three quarters immediately before June 12, 2015 ($Stockholdings_Before_MktCrash$), natural logarithm of fund age ($Ln(Age)$), and fund management fee ($Fund Fee$). Cohort fixed effects and year-quarter fixed effects are included, but for brevity, their regression coefficients are omitted from reporting. Robust standard errors (in parentheses) are clustered at the fund level. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

equations (26) and (21) of Huang and Stoll (1997) using seemingly unrelated regressions. We then separately examine the impact of deleveraging on each component. The RDD regression results are reported in Table 7.

We find that the increase in effective spread for treated (deleveraging) funds is mainly concentrated in the inventory holding component—the proportion of the spread's inventory component increases by 31%–33% after the fund crossing the deleveraging threshold. This finding is consistent with the notion that market makers are asking for a larger compensation for their inventory holdings when investor demand of the deleveraging funds' B-units declines. By contrast, the proportion of the adverse selection component does not change and the proportion of the order processing component decreases, after the fund crossing the deleveraging threshold. The finding of an insignificant change in the proportion of the adverse selection component does not support the alternative

explanation that the forced deleveraging triggers the attention of market makers, who become concerned about the quality of the underlying assets of these leveraged products and charge larger ask-bid spreads in fear of adverse selection.

Our empirical evidence hence suggests that these alternative explanations cannot account for our findings, and it is the expected subsequent deleveraging (and the associated loss of investors' interests) that drives the reduction in the market liquidity of equity units after the fund crosses the deleveraging threshold.

5.7. Forced Releveraging Placebo RDD Tests

Structured funds not only have forced deleveraging but also have the policy of forced releveraging. Deleveraging is triggered once the net asset value of tranche B equity unit falls below a certain threshold, while releveraging is triggered once the net asset value of tranche B equity unit rises above a certain

Table 9. Dynamic Treatment Effect Test for Nonparallel Trends

	Fund Flow	Stock Holdings Change	Cash Holdings Change	Quarterly Fund Return
$Treated_{ic} \times PreEventQtr2_{ct}$	0.010 (0.246)	-0.125 (0.298)	-0.015 (0.052)	0.051 (0.205)
$Treated_{ic} \times PreEventQtr1_{ct}$	0.298 (0.263)	0.0466 (0.270)	0.113 (0.074)	-0.155 (0.202)
$Treated_{ic} \times EventQtr_{ct}$	-0.432* (0.228)	-0.585** (0.240)	-0.069 (0.051)	-0.784*** (0.198)
$Treated_{ic} \times PostEventQtr1_{ct}$	-0.405** (0.200)	-0.460 (0.284)	-0.098* (0.058)	-0.571*** (0.204)
$Treated_{ic} \times PostEventQtr2_{ct}$	-0.170 (0.205)	-0.357 (0.241)	-0.034 (0.051)	-0.504*** (0.173)
$Treated_{ic} \times PostEventQtr3_{ct}$	-0.209 (0.196)	-0.386 (0.240)	-0.076 (0.047)	-0.315* (0.183)
Controls	Yes	Yes	Yes	Yes
Cohort fixed effects	Yes	Yes	Yes	Yes
Year-Qtr fixed effects	Yes	Yes	Yes	Yes
Observations	1,209	871	1,234	1,336
Adjusted R^2	0.133	0.429	0.136	0.569

Notes. This table reports the results of difference-in-differences regressions that investigate the impact of deleveraging on fund flows, stock holdings changes, cash holdings changes, and fund returns in subsequent quarters. We use three quarters before and three quarters after the deleveraging-event quarter and further include the deleveraging-event quarter to construct cohorts, and then stack all cohorts into a sample. Because we have seven quarters for each cohort, we construct six indicator variables, $PreEventQtr2_{ct}$, $PreEventQtr1_{ct}$, $EventQtr_{ct}$, $PostEventQtr1_{ct}$, $PostEventQtr2_{ct}$, $PostEventQtr3_{ct}$, to indicate the six quarters around the deleveraging event (i.e., the two quarters before the event quarter, the event quarter, and the three quarters after the event quarter, respectively). We then interact these indicator variables with $Treated_{ic}$ in dynamic treatment effect tests. Our main independent variables of interest are the interaction terms. Component terms, control variables (as in Table 8), cohort fixed effects, and year-quarter fixed effects are included, but for brevity, their regression coefficients are omitted from reporting. Robust standard errors (in parentheses) are clustered at the fund level. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

threshold. Releveraging often happens during market booming periods when the net asset value of tranche B equity unit rises quickly. Most structured funds specify the releveraging threshold net asset value of tranche B equity unit to be ¥1.50. When the net asset value of tranche B equity unit rises above this threshold (and thus fund leverage ratio drops below 1.67), the fund will reduce the net asset value of B unit to ¥1.00 and convert the exceeding amount of B unit value into equivalent tranche M parent units (and thus the fund leverage ratio increases to the initial leverage ratio of two). During our sample period, we identify 44 structured funds that were forced to releverage to

the initial leverage level.³⁰ These releveraging events provide us the opportunity to conduct another placebo test. As fund leverage are forced to go up after releveraging, we expect to see a positive impact on the market liquidity of tranche B equity units because of investor preference on high-embedded-leverage securities. However, because releveraging only causes a small increase in fund leverage (e.g., from 1.67 to 2), we expect the impact of releveraging on tranche B market to be much weaker in magnitude than that of deleveraging. The RDD results on releveraging are reported in Table A.6 in the Appendix. We find that the impact of releveraging on tranche B market liquidity is mostly

Table 10. PSM DID Analysis

Panel A: Probit regression of being treated (treated in quarter t ; yes = 1)	
$Avg. Alpha_{ict-1}$	-67.99*** (16.04)
$Ret Nav_{ict-1}$	-1.654*** (0.410)
$Ln(Size)_{ict-1}$	0.150** (0.063)
$Ln(Family Size)_{ict-1}$	-0.035 (0.099)
$LD_Before_MktCrash_{ic}$	-0.323*** (0.100)
$Stockholdings_Before_MktCrash_{ic}$	1.538*** (0.307)
Observations	1346
Pseudo- R^2	0.280

Table 10. (Continued)

Panel B: Prematching differences in fund characteristics before deleveraging					
Variable	Treatment	Control	Difference	<i>t</i> statistic	<i>p</i> value
<i>Avg. Alpha</i> _{<i>ict</i>-1}	-0.0008	0.0008	-0.0016	-1.961	0.050
<i>Ret Nav</i> _{<i>ict</i>-1}	0.013	0.037	-0.024	-1.862	0.063
<i>Ln(Size)</i> _{<i>ict</i>-1}	19.77	20.15	-0.380	-2.310	0.021
<i>Ln(Family Size)</i> _{<i>ict</i>-1}	24.71	24.54	0.168	1.185	0.237
<i>LD_Before_MktCrash</i> _{<i>ic</i>}	3.014	3.841	-0.827	7.122	0.000
<i>Stockholdings_Before_MktCrash</i> _{<i>ic</i>}	0.810	0.300	0.510	11.41	0.000
Panel C: Postmatching differences in fund characteristics before deleveraging					
Variable	Treatment	Control	Difference	<i>t</i> statistic	<i>p</i> value
<i>Avg. Alpha</i> _{<i>ict</i>-1}	0.0005	0.0008	-0.0003	-0.516	0.606
<i>Ret Nav</i> _{<i>ict</i>-1}	0.056	0.048	0.008	0.751	0.453
<i>Ln(Size)</i> _{<i>ict</i>-1}	19.93	19.58	0.350	1.278	0.202
<i>Ln(Family Size)</i> _{<i>ict</i>-1}	24.57	24.50	0.070	0.435	0.664
<i>LD_Before_MktCrash</i> _{<i>ic</i>}	3.007	3.182	-0.175	-1.110	0.268
<i>Stockholdings_Before_MktCrash</i> _{<i>ic</i>}	0.811	0.765	0.046	1.206	0.229
Panel D: DID regressions					
	<i>Fund Flow</i>	<i>Stock Holdings Change</i>	<i>Cash Holdings Change</i>	<i>Quarterly Fund Return</i>	
<i>Treated</i> _{<i>ic</i>} × <i>Post</i> _{<i>ct</i>}	-0.377*** (0.103)	-0.292** (0.117)	-0.091*** (0.025)	-0.300*** (0.067)	
<i>Post</i> _{<i>ct</i>}	0.128 (0.156)	0.245 (0.172)	-0.029 (0.044)	0.079 (0.124)	
<i>Treated</i> _{<i>ic</i>}	0.238*** (0.090)	0.252** (0.097)	0.056** (0.021)	0.122*** (0.045)	
Controls	Yes	Yes	Yes	Yes	
Cohort fixed effects	Yes	Yes	Yes	Yes	
Year-Qtr fixed effects	Yes	Yes	Yes	Yes	
Observations	460	429	497	504	
Adjusted <i>R</i> ²	0.146	0.353	0.153	0.531	

Notes. This table reports the results of PSM DID analysis that investigate the impact of deleveraging on fund flows, stock holdings changes, cash holdings changes, and fund returns in subsequent quarters. We use PSM to select the control group (i.e., structured funds that did not deleverage during the sample period) that has no observable difference in ex ante fund characteristics compared with the treatment group. Specifically, we run a probit model using our fund-quarter stacked cohort sample (that we use to estimate the DID results in Table 8). The dependent variable of the probit model equals one if fund *i* in cohort *c* in quarter *t* has been treated (i.e., has already experienced deleveraging) and equals zero otherwise. The independent variables include the lagged average fund alpha (*Avg Alpha*), lagged return on net asset value (*Ret Nav*), lagged natural logarithm of fund AUM (*Ln(Size)*), lagged natural logarithm of fund family AUM (*Ln(Family Size)*), average distance of the fund's actual leverage to its deleveraging threshold in the 90 trading days immediately before June 12, 2015 (*LD_Before_MktCrash*), and the fund's average total value of quarter-end stock holdings scaled by total assets under management in the three quarters immediately before June 12, 2015 (*Stockholdings_Before_MktCrash*). Next, we calculate the propensity score for each fund-quarter (i.e., the likelihood that fund *i* in cohort *c* in quarter *t* has been treated). Then, for each fund *i* in cohort *c*, we calculate its average propensity score for the three quarters prior to the deleveraging-event quarter. Next, within each cohort *c*, we calculate the difference in the average propensity score between each control fund and the treated fund. We then select control funds with a score difference smaller than 0.01 to reconstruct the stacked cohort sample for our difference-in-differences analysis. Panel A shows the probit regression results. Panels B and C show the differences in fund characteristics between the treatment and control funds before and after the PSM, respectively. We also report the results from two-sample *t*-tests of the differences. Panel D reports the difference-in-differences regression results using the PSM cohort sample. The dependent variable is *Fund Flow*, *Stock Holdings Change*, *Cash Holdings Change*, or *Quarterly Fund Return*. Our main independent variable of interest is *Treated* × *Post*. *Treated* equals one if fund *i* in cohort *c* is the treatment fund and equals zero otherwise. *Post* equals one if quarter *t* is a quarter after the deleveraging event in cohort *c* and equals zero otherwise. Control variables (as in Table 8), cohort fixed effects and year-quarter fixed effects are included, but for brevity, their regression coefficients are omitted from reporting. Robust standard errors (in parentheses) are clustered at the fund level. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

insignificant. However, we do find some evidence of a positive treatment effect on tranche B units' trading volume, suggesting that releveraging may increase investors' interests and activeness in trading tranche B equity units.

To summarize, the RDD estimation results in this section clearly show that deleveraging has a large and negative effect on the market liquidity of the treatment funds' tranche B equity units. This strongly supports Hypothesis 1. To the best of our knowledge, this finding

Table 11. Is the Impact on Funding Liquidity Channeled Through the Deterioration of the Market Liquidity of Equity Units?

	Fund Flow			Stock Holdings Change		
$\Delta OI_{ic} \times Post_{ct}$	0.171** (0.081)			0.154** (0.071)		
$\Delta ES_{ic} \times Post_{ct}$		−0.002** (0.001)			−0.002** (0.001)	
$\Delta Ln(Vol)_{ic} \times Post_{ct}$			0.151*** (0.047)			0.075 (0.052)
Observations	431	431	431	399	399	399
Adjusted R^2	0.121	0.124	0.155	0.362	0.362	0.364
	Cash Holdings Change			Quarterly Fund Return		
$\Delta OI_{ic} \times Post_{ct}$	0.042** (0.017)			0.152*** (0.046)		
$\Delta ES_{ic} \times Post_{ct}$		−0.001*** (0.0002)			−0.002*** (0.0005)	
$\Delta Ln(Vol)_{ic} \times Post_{ct}$			0.039*** (0.011)			0.089*** (0.033)
Observations	463	463	463	469	469	469
Adjusted R^2	0.151	0.154	0.178	0.528	0.531	0.530
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Cohort fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-Qtr fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table reports the results of OLS regressions that investigate whether larger reduction in market liquidity of equity units around the deleveraging event leads to greater decreases in fund flows, stock holdings, cash holdings, and fund performance in subsequent quarters. Based on the PSM cohort sample, we first estimate the changes in the market liquidity of equity units around the deleveraging event, ΔOI , ΔES , and $\Delta Ln(Vol)$. Specifically, to capture the change, for fund i in cohort c , we use the difference between the fund's equity units' average market liquidity (i.e., OI , ES , or $Ln(Vol)$) in the three days after the deleveraging-event day and its average market liquidity in the three days prior to the deleveraging-event day. We then use these market liquidity change variables to replace $Treated$ in Equation (5). Our main variables of interest are the interaction terms, $\Delta OI \times Post$, $\Delta ES \times Post$, and $\Delta Ln(Vol) \times Post$. $Post$ equals one if quarter t is a quarter after the deleveraging event in cohort c and equals zero otherwise. Component terms, control variables (as in Table 8), cohort fixed effects, and year-quarter fixed effects are included, but for brevity their regression coefficients are omitted from reporting. Robust standard errors (in parentheses) are clustered at the fund level. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

of a (potentially) causal effect of deleveraging on the market liquidity of high-embedded-leverage securities is novel.

6. The Impact of Deleveraging on Funding Liquidity

We conjecture that the significantly deteriorated market liquidity of equity units because of deleveraging can result in investors redeeming their investment from the treatment funds. Thus, the treatment funds likely experience large capital outflows and may fall into funding liquidity crises. To meet the significantly increased redemption demand, these funds likely need to significantly draw down their cash holdings and may have a fire sale of their stock holdings, which can reduce fund performance (Shleifer and Vishny 1992, Brunnermeier and Pedersen 2009, Acharya and Viswanathan 2011).

We use data on quarterly fund flows, portfolio holdings, and cash holdings surrounding the deleveraging events to test Hypothesis 2 (i.e., compared with control

funds, treatment funds subsequently experience large fund outflows and large reductions in stock holdings, cash holdings, and fund performance). To identify the impact of deleveraging, we use the event-window cohort approach in a DID regression framework, as suggested by Gormley and Matsa (2011).

Specifically, for each deleveraging event, we form a cohort of the deleveraging fund with all structured funds that never deleveraged during the entire sample period. We use three quarters before and three quarters after the deleveraging-event quarter (the deleveraging-event quarter is excluded from estimation) and then stack all cohorts into one sample for our DID analysis. The DID regression specification is as follows:

$$Outcome_{ict} = \alpha + \beta_1 Treated_{ic} \\ \times Post_{ct} + \beta_2 Post_{ct} + \beta_3 Treated_{ic} + \gamma X + \varepsilon_{ict}. \quad (5)$$

In Equation (5), the dependent variable, *Outcome*, is *Fund Flow*, *Stock Holdings Change*, *Cash Holdings Change*, or *Quarterly Fund Return*. *Treated* equals one if fund i in

cohort c is the treatment fund and equals zero otherwise. $Post$ equals one if quarter t is a quarter after the deleveraging event in cohort c and equals zero otherwise. Our variable of interest is $Treated \times Post$. We expect its coefficient, β_1 , to be significantly negative for DID regressions using the aforementioned outcome variables if the treatment funds experience funding liquidity crises subsequent to deleveraging. X is a vector of control variables.

Following the literature (Chen et al. 2010, Kacperczyk and Schnabl 2013), we include as controls the lagged average fund alpha ($Avg. Alpha$), lagged return on net asset value ($Ret Nav$), lagged natural logarithm of fund AUM ($Ln(Size)$), lagged natural logarithm of fund family AUM ($Ln(Family Size)$), the average distance of the fund's actual leverage to its deleveraging threshold in the 90 trading days immediately before June 12, 2015 ($LD_Before_MktCrash$), the fund's average total value of quarter-end stock holdings scaled by total assets under management in the three quarters immediately before June 12, 2015 ($Stockholdings_Before_MktCrash$), fund management fee ($Fund Fee$), and natural logarithm of fund age ($Ln(Age)$). We further control for cohort fixed effects and year-quarter fixed effects, and cluster standard errors at the fund level. Table 8 reports the DID regression results.

As expected, the regression coefficient of $Treated \times Post$ is significantly negative at the 1% level across all regressions with different dependent variables. The results clearly show that compared with structured funds that did not deleverage, the treatment funds saw large redemption outflows and large reductions in stock holdings, cash holdings, and fund returns in subsequent quarters. These results indicate that these funds experienced investor runs and severe funding liquidity crises. In terms of economic magnitude, relative to funds that did not deleverage, treatment funds experienced a large ex post reduction in fund flows (0.343) equivalent to 87% of the sample standard deviation of fund flows, a large incremental reduction in fund stock holdings (0.358) equivalent to 74% of the sample standard deviation of stock holdings, a large incremental decrease in fund cash holdings (0.104) equivalent to 80% of the sample standard deviation of cash holdings, and a large incremental decrease in quarterly fund returns (0.397) equivalent to 114% of the sample standard deviation of fund returns.

An important premise of DID analysis is the parallel-trend condition: the outcome variable of the treatment and control groups should not be trending differently before the onset of treatment. To test whether the parallel-trend condition is satisfied, we use three quarters before and three quarters after the deleveraging-event quarter and further include the deleveraging-event

quarter to construct cohorts, and then stack all cohorts into one sample. Because we have seven quarters for each cohort, we construct six indicator variables— $PreEventQtr2$, $PreEventQtr1$, $EventQtr$, $PostEventQtr1$, $PostEventQtr2$, and $PostEventQtr3$ —to indicate the six quarters around the deleveraging event (i.e., the two quarters before the event quarter, the event quarter, and the three quarters after the event quarter, respectively). We then interact these indicator variables with $Treated$ in dynamic DID regressions. Our main variables of interest are the interaction terms. Component terms, control variables (as in Table 8), cohort fixed effects, and year-quarter fixed effects are also included, but for brevity, they are omitted from reporting. Table 9 reports the results.

From Table 9, it is clear that the coefficients of the interaction terms, $PreEventQtr2 \times Treated$ and $PreEventQtr1 \times Treated$, are statistically insignificant across all regressions with different dependent variables. The biggest treatment effects occur in the event quarter and in the first quarter following; the effects then decay gradually from the second quarter onward. Thus, the results are consistent with deleveraging having a causal impact on subsequent fund flows, stock holdings, cash holdings, and fund returns. There is no nonparallel trend in any of these outcome variables before the onset of treatments.

As the difference-in-differences OLS results in Table 8 only summarize the average treatment effects on the outcome variables, we further estimate Equation (5) using quantile regressions to study the relationships at different points in the conditional distributions of the outcome variables. We examine the conditional relationships at the quantiles 10, 20, 30, 50, 70, and 90 of the outcome variables. Table A.7 reports the results using bootstrapped standard errors.

We find that the coefficient of $Treated \times Post$ is significantly negative in most of the quantile regressions. Moreover, we find that the higher the quantiles of the outcome variables, the larger the treatment effects of deleveraging on the outcome variables. For higher quantiles of the outcome variables with positive quantile values (in brackets), the coefficients of $Treated \times Post$ are very large in magnitude compared with the respective quantile values. Thus, deleveraging exerts a large impact that causes fund flows, stock holdings changes, cash holdings changes, and fund returns of the treatment funds to become negative even at higher quantile values of these outcome variables. This finding suggests that these funds fall into funding liquidity crises.

There could be an endogeneity concern for the aforementioned DID analysis because the treatment (i.e., deleveraging) may correlate with recent fund performance and/or other fund characteristics. To alleviate this endogeneity concern, we follow Fang et al.

(2014) and Brogaard et al. (2017): we use PSM to select the control group (i.e., fund-quarter observations of structured funds that did not deleverage during the sample period) that has no observable difference in ex ante fund characteristics compared with the treatment group. Specifically, we first run a probit model using our fund-quarter stacked cohort sample (that we use to estimate the DID results in Table 8).

The dependent variable of the probit model equals one if fund i in cohort c in quarter t has been treated (i.e., has already experienced deleveraging) and equals zero otherwise. The independent variables include the lagged average fund alpha (*Avg. Alpha*), lagged return on net asset value (*Ret Nav*), lagged natural logarithm of fund AUM ($\ln(\text{Size})$), lagged natural logarithm of fund family AUM ($\ln(\text{Family Size})$), average distance of the fund's actual leverage to its deleveraging threshold in the 90 trading days immediately before June 12, 2015 (*LD_Before_MktCrash*), and the fund's average total value of quarter-end stock holdings scaled by total assets under management in the three quarters immediately before June 12, 2015 (*Stockholdings_Before_MktCrash*). Next, we calculate the propensity score for each fund-quarter (i.e., the likelihood that fund i in cohort c in quarter t has been treated). Then, for each fund i in cohort c , we calculate its average propensity score in the three quarters before the deleveraging-event quarter. Next, within each cohort c , we calculate the difference in the average propensity score between each control fund (i.e., structured funds that did not deleverage during the entire sample period) and the treatment fund. We then select control funds with a score difference smaller than 0.01 to reconstruct the stacked cohort sample for our DID analysis. Table 10 reports the results.

Panel A of Table 10 shows the probit regression results. As expected, funds with higher lagged alpha and/or higher lagged return on net asset value are less likely to deleverage in the next quarter. Moreover, larger funds and funds with shorter average distance of actual leverage to its deleveraging threshold and/or higher average stock holdings before the market crash are more likely to subsequently deleverage. Panels B and C show the differences in fund characteristics between the treatment and control funds before and after the PSM, respectively. Two-sample t tests show that before PSM, the treatment funds have significantly worse performance, smaller fund size, shorter leverage distance to threshold, and higher stock holdings than the control funds (Panel B). After PSM, there is no significant difference in any of the fund characteristics between the treatment and control funds (Panel C), which suggests that the matching process has successfully removed the fund-characteristics heterogeneity.

Panel D reports the DID regression results with the PSM cohort sample. It is clear that the results with the

PSM are qualitatively very similar to those without PSM. We further exclude all structured funds with leverage limits not being five from the sample; the DID regression results, reported in Table A8 in the Appendix, are qualitatively similar to those in Panel D of Table 10.

These results confirm that deleveraging results in large fund outflows and large decreases in stock holdings, cash holdings, and performance for the treatment funds. These findings strongly support Hypothesis 2.

7. Market Liquidity Deterioration and Funding Liquidity Crises

The funding liquidity crises of the treatment funds are likely caused by the large reduction in market liquidity of their tranche B equity units after deleveraging. The larger the reduction in market liquidity, the greater the incentive for investors to redeem their investment from the treatment funds rather than to sell to the market their fund units at discounted prices. This leads to larger redemption outflows and hence larger reduction in stock holdings, cash holdings, and fund performance. In this section, we test Hypothesis 3: the impact of deleveraging on fund flows, stock holdings, cash holdings, and fund returns is channeled through the deterioration of the market liquidity of the equity units.

Based on the aforementioned PSM cohort sample, we first estimate the changes in the equity units' market liquidity around the deleveraging-event day, ΔOI_{ic} , ΔES_{ic} , and $\Delta \ln(Vol)_{ic}$. Specifically, to capture the change, for fund i in cohort c , we use the difference between the average market liquidity of the fund's equity units (i.e., OI , ES , or $\ln(Vol)$) in the three days after the deleveraging-event day and its average market liquidity in the three days prior to the deleveraging-event day. We then use these market liquidity change variables to replace $Treated_{ic}$ in Equation (5). We expect greater reduction in equity units' market liquidity around the deleveraging event to result in greater decreases in fund flows, stock holdings, cash holdings, and fund performance in subsequent quarters. Thus, our main variables of interest are $\Delta OI \times Post$, $\Delta ES \times Post$, and $\Delta \ln(Vol) \times Post$, and we expect the coefficients of $\Delta OI \times Post$ and $\Delta \ln(Vol) \times Post$ (the coefficient of $\Delta ES \times Post$) to be significantly positive (negative). Component terms, control variables, cohort fixed effects, and year-quarter fixed effects are also included but for brevity are not reported. Table 11 reports the results.

As expected, the coefficients of $\Delta OI \times Post$ and $\Delta \ln(Vol) \times Post$ (the coefficient of $\Delta ES \times Post$) are all positive (all negative) across different regressions using different outcome variables as the dependent variables. Moreover, the interaction term coefficients are statistically significant at least at the 5% level in 11 of the 12 regressions. The results clearly show that larger reduction in the market liquidity of equity units

around the deleveraging event leads to greater decreases in fund flows, stock holdings, cash holdings, and fund performance in subsequent quarters. This result supports Hypothesis 3: the impact of deleveraging on fund flows, stock holdings, cash holdings, and fund returns is channeled through the deterioration of equity units' market liquidity.

8. Conclusion

Leverage-constrained investors prefer to invest in high-embedded-leverage securities (Frazzini and Pedersen 2014). If issuers of such securities are forced to deleverage, deleveraging will make these securities unattractive to leverage-constrained investors, which may result in a reduction in investor demand and thus significant deterioration of the market liquidity of these securities. The deterioration of market liquidity can in turn tighten the funding constraints of the security issuers, which can potentially lead to funding liquidity crises. In this paper, we use the forced deleveraging of structured mutual funds during the 2015 Chinese stock market crash to study the impact of deleveraging on market liquidity, funding constraints, and performance of the treatment funds.

Our RDD analysis shows that deleveraging exerts a large and negative impact on the market liquidity of the treatment funds' equity units. When a fund just crosses the deleveraging spread threshold (and thus is expected to deleverage soon), its order imbalance decreases on average by 2.1 times, effective spread increases by 5.8 times, and trading volume decreases by 43%, relative to their respective sample standard deviations.

Our DID analysis further shows that, compared with PSM-matched structured funds that did not deleverage, the treatment funds saw large redemption fund outflows and large reductions in stock holdings, cash holdings, and fund performance in subsequent quarters, indicating that these funds experienced funding liquidity crises. Moreover, we show that larger reduction in market liquidity of equity units around the deleveraging event leads to greater decreases in fund flows, stock holdings, cash holdings, and fund performance in subsequent quarters. Thus, the impact of deleveraging on fund flows, stock holdings, cash holdings, and fund

returns is channeled through the deterioration of the market liquidity of equity units.

This study contributes to the literature by documenting novel evidence of a negative and (potentially) causal impact of deleveraging on the market liquidity of high-embedded-leverage securities. We further show how this negative impact on market liquidity can translate into funding liquidity crises for financial institutions that issue such securities. This novel link between deleveraging, market liquidity of high-embedded-leverage securities, and funding liquidity of financial institutions issuing such securities is distinct from the known interaction between traders' funding liquidity and asset market liquidity (Brunnermeier and Pedersen 2009). Given that financial institutions around the world currently have very high levels of leverage because of the extremely low interest rates in the past decade (as a result of the quantitative-easing programs of central banks after the global financial crisis and more recently during the COVID-19 crisis), our findings may be especially relevant to policy makers, academics, and practitioners who are concerned about the impact of deleveraging.

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Appendix

Table A.1. Variable Description

Variable	Frequency	Definition
Leverage	Daily	Total assets under management of the fund / total net asset value of its tranche B equity units.
OI	Daily	$OI_{it} = \frac{Buy\ orders_{it} - Sell\ orders_{it}}{0.5 \times (Buy\ orders_{it} + Sell\ orders_{it})}$. We use the tick-by-tick trading data of tranche B equity units to calculate the sum of all buy orders and the sum of all sell orders for each structured fund in each trading day. OI is then calculated for each fund-day using this equation.

Table A.1. (Continued)

Variable	Frequency	Definition
ES	Daily	$ES_{it} = \frac{100 \times (\text{Transaction Price}_{it} - 0.5 \times (\text{Bid}_{it} + \text{Ask}_{it})) \times 2}{0.5 \times (\text{Bid}_{it} + \text{Ask}_{it})}$. To calculate ES for a fund's equity units in a trading day, we first calculate the effective spread of each trade during the day using this equation and then average the effective spread across the day for that fund.
$\ln(\text{Vol})$	Daily	Natural logarithm of the daily trading volume of the fund's tranche B equity units.
Turnover	Daily	Daily trading volume divided by the total number of tranche B equity units outstanding for the fund.
Fund Return	Daily	Daily return (with dividend reinvestment) of the fund's tranche B equity units.
Stdret	Daily	Standard deviation of 5-minute-interval fund returns during the day. We first use tick-by-tick trading data of the fund's tranche B equity units to calculate 5-minute-interval fund returns. We then calculate the standard deviation of 5-minute returns during the day.
Alpha	Daily	Daily 3-factor alpha of the fund's tranche B equity units estimated using the Fama and French (1996) 3-factor model and the daily MKT, SMB, and HML factors constructed for the Chinese A-share market.
LD_Before_MktCrash	Daily	The distance of the fund's actual leverage to its deleveraging threshold in the 90 trading days immediately before June 12, 2015
Premium	Daily	(Closing price per tranche B unit – net asset value per tranche) / net asset value per tranche B unit
High Prem	Daily	An indicator variable which equals 1 if a fund's short-run average tranche B premium over the past five trading days is greater than the median short-run average premium of all fund-day observations in the left side of the deleveraging spread threshold in the respective RDD regression sample and equals 0 otherwise.
Fund Flow	Quarterly	(The number of newly issued tranche A and B units in the quarter – the number of redeemed tranche A and B units in the quarter) / the total number of tranche A and B units at the beginning of the quarter.
Stock Holdings Change	Quarterly	(The total value of stock holdings at the end of the quarter – the total value of stock holdings at the end of the last quarter $\times (1 + \text{Ret Nav})$) / the total value of stock holdings at the end of the last quarter.
Cash Holdings Change	Quarterly	(The total value of cash and cash equivalents at the end of the quarter – the total value of cash and cash equivalents at the end of the last quarter) / the total value of cash and cash equivalents at the end of the last quarter.
Quarterly Fund Return	Quarterly	Quarterly return (with dividend reinvestment) of the fund's tranche B equity units calculated as $\prod_{j=1}^T (1 + \text{Fund Return}_j)$, with j indicating trading day j of the quarter and T being the number of trading days of the quarter.
Avg. Alpha	Quarterly	Average daily fund 3-factor alpha across all trading days in a quarter.
Ret Nav	Quarterly	Net profit of the fund in the quarter / total assets under management of the fund at the end of the quarter.
$\ln(\text{Size})$	Quarterly	Natural logarithm of total assets under management of the fund.
$\ln(\text{Family Size})$	Quarterly	Natural logarithm of total assets under management of the fund's family.
Stockholdings_Before_MktCrash	Quarterly	The fund's total value of quarter-end stock holdings scaled by total assets under management in the three quarters immediately before June 12, 2015
Fund Fee	Time-invariant	Fund management fee as a percentage of total assets under management.
Fund Age	Time-invariant	The number of years from the year of the fund's establishment to 2017.
$\ln(\text{Age})$	Time-invariant	Natural logarithm of fund's age.

Table A.2. Leverage Threshold Distribution of the Structured Funds

Leverage threshold	Frequency	Percent
3.00	1	0.55
3.22	4	2.19
3.50	3	1.64
3.86	1	0.55
4.33	1	0.55
4.70	1	0.55
5.00	129	70.49
6.00	6	3.28
7.67	37	20.22
Total	183	100

Note. This table reports the leverage threshold distribution of the structured fund sample according to the threshold net asset value of tranche B equity unit that triggers deleveraging.

Table A.3. Treatment Effect Under Local Randomization Inference

	Treatment effect
<i>OI</i>	−0.633***
<i>p</i> value	(0.004)
95% confidence interval	[−1.097, −0.154]
<i>ES</i>	59.94***
<i>p</i> value	(0.002)
95% confidence interval	[18.95, 99.84]
<i>Ln(Vol)</i>	−1.469***
<i>p</i> value	(0.008)
95% confidence interval	[−2.601, −0.391]
<i>Turnover</i>	−0.398**
<i>p</i> value	(0.022)
95% confidence interval	[−0.778, −0.061]
Bandwidth/window	[−0.015, 0.015]
Number of treated observations	31
Number of control observations	44

Notes. This table reports the treatment effect of deleveraging on tranche B's market liquidity under small-sample local randomization inference. We follow the approach of Cattaneo et al. (2016) by selecting a narrow window near the deleveraging spread threshold (i.e., deleveraging spread = 0), where the local randomization assumption is most plausible, and we use their local randomization inference method to estimate the local average treatment effect. The market liquidity variable is *OI*, *ES*, *Ln(Vol)*, or *Turnover*. This table provides the point estimate of the treatment effect, the bootstrapped *p* value (in parentheses) under the null hypothesis that the treatment effect is 0, and the 95% confidence interval (in brackets) of the treatment effect. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.4. Impact of Deleveraging on Tranche B Market Liquidity (Leverage Limit = 5)

	<i>OI</i>		<i>ES</i>		<i>Ln(Vol)</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated_{it}</i>	−1.161*** (0.132)	−0.712*** (0.164)	114.1*** (13.86)	54.05*** (13.97)	−1.350*** (0.442)	−1.158*** (0.179)
Controls	No	Yes	No	Yes	No	Yes
Bandwidth	[−0.077, 0.077]	[−0.077, 0.077]	[−0.071, 0.071]	[−0.071, 0.071]	[−0.114, 0.114]	[−0.114, 0.114]
Observations	634	315	533	261	1,320	570
Adjusted <i>R</i> ²	0.236	0.482	0.279	0.629	0.031	0.862

Notes. This table reports the results of RDD regressions that investigate the impact of deleveraging on tranche B's market liquidity. We include only those funds with their leverage limit being five. The dependent variable is *OI*, *ES*, or *Ln(Vol)*. The main independent variable of interest is *Treated*, which equals one if deleveraging spread of a fund-day is greater than the threshold value of zero (and hence the fund needs to subsequently deleverage) and equals zero otherwise. Control variables include the lagged dependent variable, lagged fund alpha (*Alpha*), lagged natural logarithm of fund AUM (*Ln(Size)*), lagged natural logarithm of fund family AUM (*Ln(Family Size)*), average distance of the fund's actual leverage to its deleveraging threshold in the 90 trading days immediately before June 12, 2015 (*LD_Before_MktCrash*), the fund's average total value of quarter-end stock holdings scaled by total assets under management in the three quarters immediately before June 12, 2015 (*Stockholdings_Before_MktCrash*), fund management fee (*Fund Fee*), and natural logarithm of fund age (*Ln(Age)*). We follow Calonico et al.'s (2017) data-driven bandwidth selection approach and select one common bandwidth on both sides of the threshold. The selection of bandwidth is reported in brackets. Robust standard errors (in parentheses) are clustered at both the fund and year-day levels. Control variables (as in Table 4) are included, but for brevity, their regression coefficients are omitted from reporting. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.5. Testing the Alternative Trading-Premium Explanation

	<i>OI</i>	<i>ES</i>	<i>Ln(Vol)</i>
<i>Treated_{it}</i>	−0.788*** (0.220)	86.45*** (16.40)	−1.410*** (0.345)
<i>High Prem_{it}</i>	−0.328** (0.153)	29.02** (12.82)	−0.119 (0.154)
<i>Treated_{it} × High Prem_{it}</i>	0.189 (0.308)	−53.37** (25.77)	0.482 (0.319)
Controls	Yes	Yes	Yes
Bandwidth	[−0.077, 0.077]	[−0.071, 0.071]	[−0.114, 0.114]
Observations	339	283	615
Adjusted <i>R</i> ²	0.424	0.593	0.870

Notes. The table reports the results of RDD regressions that investigate the impact of deleveraging on tranche B market liquidity. The RDD regression sample includes all fund-day observations with deleveraging spread (i.e., the forcing variable) close to the threshold of zero. The dependant variable is *OI*, *ES*, or *Ln(Vol)*. The main independent variable of interest is *Treated × High Prem*. *Treated* is defined as in Table 4. *High Prem* equals one if a fund's short-run average tranche B premium (i.e., (price per unit – net asset value per unit)/net asset value per unit) over the past five trading days is greater than the median short-run average premium of all fund-day observations in the left side of the deleveraging spread threshold in the respective RDD regression sample and equals zero otherwise. We follow the date-driven bandwidth selection approach of Calonico et al. (2017) and select one common bandwidth on both sides of the threshold. Robust standard errors (in parentheses) are clustered at both the fund and year-day levels. Control variables (as in Table 4) are included, but for brevity, their regression coefficients are omitted from reporting. Detailed description of the variables is presented in the Table A.1.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.6. Placebo RDD Test on the Impact of Releveraging on Tranche B Market Liquidity

	<i>OI</i>		<i>ES</i>		<i>Ln(Vol)</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treated_{it}</i>	−0.010 (0.037)	0.048 (0.061)	−0.306 (0.942)	2.152 (2.128)	0.202*** (0.038)	0.083 (0.074)
<i>OI_{it-1}</i>	0.279*** (0.048)	0.186* (0.099)				
<i>ES_{it-1}</i>			0.673*** (0.103)	0.242 (0.160)		
<i>Ln(Vol)_{it-1}</i>					0.896*** (0.018)	0.870*** (0.049)
<i>Turnover_{it-1}</i>						
<i>Alpha_{it-1}</i>	0.989*** (0.325)	1.730** (0.807)	4.937 (23.61)	3.481 (29.56)	−0.143 (0.602)	−2.066** (1.022)
<i>Ln(Size)_{it-1}</i>	0.018*** (0.005)	0.020 (0.016)	−0.752* (0.426)	0.307 (0.711)	0.096*** (0.024)	0.077 (0.048)
<i>Ln(Family Size)_{it-1}</i>	0.004 (0.007)	−0.044 (0.032)	1.153 (0.826)	−1.668 (2.267)	−0.008 (0.019)	0.009 (0.040)
<i>LD_Before_MktCrash_i</i>	0.019 (0.015)	−0.084 (0.069)	−1.173 (3.595)	−0.301 (6.381)	−0.014 (0.056)	−0.054 (0.080)
<i>Stockholdings_Before_MktCrash_i</i>	−0.069* (0.039)	0.452 (0.284)	1.564 (12.41)	−7.612 (31.15)	0.143 (0.201)	0.432* (0.229)
<i>Fund Fee_i</i>	−0.048 (0.044)	−0.130 (0.247)	−6.846* (3.763)	−22.27 (20.07)	−0.118 (0.071)	0.018 (0.310)
<i>Ln(Age)_i</i>	−0.056 (0.039)	0.064 (0.114)	−0.763 (1.540)	−2.565 (3.419)	0.025 (0.049)	−0.198 (0.166)
Bandwidth	[−0.308, 0.051]	[−0.038, 0.038]	[−0.198, 0.071]	[0.053, 0.053]	[−0.182, 0.037]	[−0.034, 0.034]
Observations	3,098	168	1,454	246	1,254	145
Adjusted <i>R</i> ²	0.109	0.055	0.418	0.142	0.956	0.957

Notes. This table reports the results of RDD regressions that investigate the impact of releveraging on tranche B's market liquidity. The RDD regression sample includes all fund-day observations with releveraging spread (i.e., the forcing variable, which equals the difference between the net asset value of tranche B equity unit and the releveraging threshold value of B unit) close to the threshold of zero. The dependant variable is *OI*, *ES*, or *Ln(Vol)*. The main independent variable of interest is *Treated*, which equals one if releveraging spread of a fund-day is greater than the threshold value of zero (and hence the fund needs to subsequently releverage) and equals zero otherwise. Control variables are the same as those in Table 4. We follow the data-driven bandwidth selection approach of Calonico et al. (2017) and select two distinct bandwidths on the two sides of the threshold (the odd columns) or one common bandwidth on both sides of the threshold (the even columns). The selection of bandwidth is reported in brackets. Robust standard errors (in parentheses) are clustered at both the fund and year-day levels. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.7. Impact of Deleveraging on Funding Liquidity Using Quantile Regressions

Dependent variable	Key independent variable	Quantile					
		P10	P20	P30	P50	P70	P90
<i>Fund Flow</i>	$Treated_{ic} \times Post_{ct}$	0.010 (0.097) [−0.451]	0.092 (0.100) [−0.277]	−0.028 (0.091) [−0.175]	−0.267* (0.136) [−0.045]	−0.566*** (0.141) [0.000]	−0.705* (0.359) [0.334]
	Observations	1030	1030	1030	1030	1030	1030
	Pseudo- R^2	0.145	0.130	0.100	0.047	0.052	0.200
<i>Stock Holdings Change</i>	$Treated_{ic} \times Post_{ct}$	−0.169* (0.100) [−0.406]	−0.172* (0.089) [−0.192]	−0.267*** (0.071) [−0.116]	−0.254** (0.115) [−0.007]	−0.640*** (0.205) [0.103]	−1.244*** (0.364) [0.627]
	Observations	741	741	741	741	741	741
	Pseudo- R^2	0.416	0.361	0.315	0.242	0.239	0.296
<i>Cash Holdings Change</i>	$Treated_{ic} \times Post_{ct}$	−0.021 (0.018) [−0.051]	−0.022* (0.011) [−0.030]	−0.024** (0.011) [−0.017]	−0.052*** (0.015) [−0.003]	−0.130*** (0.043) [0.008]	−0.232** (0.098) [0.065]
	Observations	1046	1046	1046	1046	1046	1046
	Pseudo- R^2	0.070	0.059	0.042	0.031	0.079	0.204
<i>Quarterly Fund Return</i>	$Treated_{ic} \times Post_{ct}$	−0.009 (0.055) [−0.373]	−0.090 (0.068) [−0.138]	−0.207*** (0.061) [−0.043]	−0.309*** (0.057) [0.018]	−0.407*** (0.047) [0.074]	−0.560*** (0.115) [0.429]
	Observations	1,142	1,142	1,142	1,142	1,142	1,142
	Pseudo- R^2	0.501	0.433	0.355	0.301	0.367	0.421
	Controls	Yes	Yes	Yes	Yes	Yes	Yes
	Cohort fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
	Year-Qtr fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table reports the results of difference-in-differences analysis that investigates the impact of deleveraging on fund flows, stock holdings changes, cash holdings changes, and fund returns in subsequent quarters. For each deleveraging event, we form a cohort of the deleveraging fund with all structured funds that did not deleverage during the entire sample period. We use three quarters before and three quarters after the deleveraging-event quarter (the deleveraging-event quarter is excluded) and then stack all cohorts into one sample for our difference-in-differences analysis. We use quantile regressions to study the relationships at different points in the conditional distributions of the dependent variable. The dependent variable is *Fund Flow*, *Stock Holdings Change*, *Cash Holdings Change*, or *Quarterly Fund Return*. We examine the conditional relationships at the quantiles 10, 20, 30, 50, 70, and 90 of the dependent variable. Our main independent variable of interest is $Treated \times Post$. $Treated$ equals one if fund i in cohort c is the treatment fund and equals zero otherwise. $Post$ equals one if quarter t is a quarter after the deleveraging event in cohort c and equals zero otherwise. Component terms, control variables (as in Table 8), cohort fixed effects, and year-quarter fixed effects are included, but for brevity, their regression coefficients are omitted from reporting. The respective quantile values of the dependent variable are reported in brackets. Bootstrapped standard errors are reported in parentheses. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.8. PSM DID Analysis (Leverage Limit = 5)

	Fund flow	Stock holdings change	Cash holdings change	Quarterly fund return
$Treated_{ic} \times Post_{ct}$	−0.326*** (0.118)	−0.250* (0.132)	−0.075*** (0.028)	−0.140** (0.058)
$Post_{ct}$	0.075 (0.174)	0.186 (0.184)	−0.027 (0.047)	0.001 (0.120)
$Treated_{ic}$	0.239** (0.101)	0.241** (0.106)	0.057** (0.024)	0.045 (0.042)
Controls	Yes	Yes	Yes	Yes
Cohort fixed effects	Yes	Yes	Yes	Yes
Year-Qtr fixed effects	Yes	Yes	Yes	Yes
Observations	321	333	357	359
Adjusted R^2	0.190	0.380	0.175	0.623

Notes. This table reports the results of propensity-score-matched difference-in-differences analysis that investigate the impact of deleveraging on fund flows, stock holdings changes, cash holdings changes, and fund returns in subsequent quarters. We include only those funds with their leverage limit being five. The dependent variable is *Fund Flow*, *Stock Holdings Change*, *Cash Holdings Change*, or *Quarterly Fund Return*. Our main independent variable of interest is $Treated \times Post$. $Treated$ equals one if fund i in cohort c is the treatment fund and equals zero otherwise. $Post$ equals one if quarter t is a quarter after the deleveraging event in cohort c and equals zero otherwise. Control variables (as in Table 8), cohort fixed effects and year-quarter fixed effects are included, but for brevity, their regression coefficients are omitted from reporting. Robust standard errors (in parentheses) are clustered at the fund level. Table A.1 presents detailed description of the variables.

*, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Endnotes

¹ For example, the debt-to-equity ratio of financial institutions in 2015 is 9.6 times in Australia, 6.0 times in the United Kingdom, 4.3 times in Germany, and 2.2 times in the United States (see <https://data.oecd.org/corporate/financial-corporations-debt-to-equity-ratio.htm#indicator-chart>).

² For instance, Barber and Odean (2000) show that individual investors tilt their common stock investment toward high beta stocks. Baker et al. (2011) and Christoffersen and Simutin (2017) suggest that institutional investors similarly prefer high beta stocks.

³ If a structured fund is classified as a bond fund (which can only invest less than 20% of its assets under management in stocks), then the limit of initial leverage is even higher at 10/3 and that of subsequent leverage is 8.

⁴ The public announcement of deleveraging will be issued on several major Chinese financial websites including <http://www.jrj.com.cn/>, <https://business.sohu.com/>, and <https://finance.sina.com.cn/>.

⁵ It is widely believed that structured funds helped amplify the Chinese stock market crash. For example, an article from an influential Chinese financial magazine, *21st Century Economic Report*, states that structured funds are to be blamed for amplifying the stock market boom and for the subsequent bust in 2014 and 2015. See <http://fund.eastmoney.com/news/1590,20150717528126136.html>.

⁶ Note that M units are only temporary and are generated during the deleveraging/releveraging processes (we discuss releveraging in Section 5). We check and find that, during the 2014–2016 sample period, among the 183 structured funds in our sample, only 52 funds ever have M units outstanding in their quarterly reports. Before June 12, 2015, only 25 funds ever have M units. After the debut of the market crash, 51 funds have M units. Over the sample period, the average ratio of quarter-end M units to all units outstanding, $M/(A + B + M)$, is only 0.04. Before June 12, 2015, that ratio is 0.003; after June 12, 2015, that ratio is 0.055. M units are held by investors of the structured funds and not by their parent companies.

⁷ When we compare market liquidity of fund-day observations of funds that cross the deleveraging threshold (and thus will be forced to deleverage in the next two trading days) with market liquidity of fund-day observations of funds that do not cross the threshold, the suitable methodology is a RDD approach because we examine market liquidity of fund-day observations just above vs. just below the threshold. After documenting that crossing the deleveraging threshold indeed significantly reduces the market liquidity of the deleveraging funds' equity units, we next examine whether the deleveraged funds, compared with the non-deleveraged structured funds, experience funding liquidity crisis subsequent to deleveraging using fund-quarter level data. Here we are comparing deleveraged funds versus non-deleveraged funds and thus we cannot use RDD anymore. To investigate whether deleveraged funds (compared with non-deleveraged funds) suffer more on fund flows and performance post the deleveraging treatment, the suitable methodology is a stacked-cohort DID approach.

⁸ This approach has been applied in the recent literature (Fang et al. 2014, Brogaard et al. 2017).

⁹ Frazzini and Pedersen (2012) find similar results for equity options, index options, and leveraged ETFs.

¹⁰ See also Acharya and Viswanathan (2011) and Greenwood et al. (2015).

¹¹ Gao et al. (2020) suggest that investor prior trading experience and trading performance extrapolation are also key determinants of the popularity of Chinese structured funds.

¹² According to the design of structure funds, investors can combine one A debt unit and one B equity unit into two parent (tranche M) units; they can then redeem the M units and receive cash from the fund.

¹³ These public announcements are available on <http://fund.eastmoney.com/> and other websites.

¹⁴ Among these 78 structured funds, five funds deleveraged more than once (four deleveraged twice and one deleveraged three times) during our sample period. Our results remain qualitatively unchanged when we exclude these five funds from the analyses.

¹⁵ Because Chinese regulatory authority, for market stability concern, stipulates that equity and fund markets cannot rise or fall by more than 10% in any single day (Tao et al. 2017), the average daily returns of the funds in our sample are relatively low compared with those from capital markets in the United States and elsewhere.

¹⁶ Fund age is time invariant in CSMAR and reflects the age of the fund when the database started to cover that fund. Fund management fee is also time invariant and reflects the fee specified in the fund's prospectus.

¹⁷ Our results do not change in any qualitative manner if we only include either lagged fund return or lagged alpha.

¹⁸ The economic magnitude of the leverage–market liquidity relation is large. A one-standard-deviation increase in leverage in the sample (i.e., 0.853) is on average related to an increase in order imbalance equivalent to 10.5% of its sample standard deviation, a decrease in effective spread equivalent to 8.2% of its sample standard deviation, an increase in trading volume equivalent to 22.0% of its sample standard deviation, and an increase in turnover equivalent to 10.7% of its sample standard deviation. The standard deviation of *OI*, *ES*, *Ln(Vol)*, and *Turnover* in the sample is 0.529, 19.031, 3.023, and 0.375, respectively.

¹⁹ The inflection point can be obtained by taking the first-order derivative of *OI* with regards to *Leverage* and setting it to zero (i.e., $8 = 0.065/(2 \times 0.004)$).

²⁰ Because deleveraging involves reducing the number of B units outstanding, our RDD analysis compares the market liquidity of the equity units of structured funds that are just below the threshold with that of structured funds just crossing the threshold before these latter funds actually deleveraged in the next two trading days.

²¹ The bin width on the left (right) side of the threshold is 0.025 (0.020). Because funds that crossed the deleveraging threshold will be forced to deleverage in the next two trading days and recalculate their tranche A and B unit numbers and net asset values, there are much fewer observations on the right-hand side of the threshold (i.e., deleveraging spread is greater than zero) than on the left-hand side of the threshold (i.e., deleveraging spread is less than zero). Thus, a few bins to the right of the threshold are empty.

²² In the current draft of the paper, for brevity concern we drop the regression results with *Turnover* as the dependent variable from the RDD and DID regression tables. The *Turnover* regression results are similar to, but weaker (i.e., sometimes insignificant) than, the reported results with *Ln(Vol)* as the dependent variable.

²³ We also find that further adding control variables of tranche A debt units does not change the results in any qualitative manner.

²⁴ We thank an anonymous referee for pointing this out to us.

²⁵ We only report the RDD estimation results using one common bandwidth on both sides of the threshold, as the results using two distinct bandwidths on the two sides of the threshold are qualitatively similar to the reported results.

²⁶ Institutional investors may be better informed than individual investors especially in the Chinese capital markets. If managerial quality signaling drives the ex post market liquidity reduction of equity units, we would expect the signal to be more important to uninformed individual investors than to the better-informed institutional investors. However, when we interact Tranche B institutional ownership with *Treated*, the interaction term is insignificant statistically across the RDD regressions. This finding does not necessarily

mean that there is no relation between ex ante institutional investor holdings and ex post reduction in equity unit market liquidity. When funds cross the deleveraging threshold, individual investors could be the marginal investor for both deleveraging funds with high institutional ownership and those with low institutional ownership. Thus, when we interact institutional ownership with *Treated*, the insignificant coefficient estimate of the interaction term may just reflect the marginal impact of individual investors.

²⁷ We also define *High Prem* using the fund's average tranche B trading premium over the past three or seven trading days, results are qualitatively unchanged.

²⁸ The finding is reasonable. Although deleveraging causally and negatively impacts the market liquidity of fund equity units because of investors' preferences for high-embedded-leverage securities, the market liquidity of tranche A debt units should not be negatively impacted. As mentioned earlier, if anything, deleveraging may have some positive effect on tranche A market liquidity for two reasons. First, because tranche A debt units have now become safer (given deleveraging), they may attract more risk-averse investors. Second, given the significant deterioration of tranche B market liquidity, instead of directly selling their B units to the market, some investors may choose to purchase A units from the market to combine with their B units into parent units and then redeem their investment from the fund.

²⁹ We thank an anonymous referee for suggesting this alternative explanation to us.

³⁰ In total these 44 structured funds have 58 releveraging events during our sample period.

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